

The Incremental Value of Player Information in Football Match Prediction

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Abstract

Closing bookmaker odds provide a benchmark for football forecasting because they summarize information available shortly before kick-off. Yet changing line-ups and fluctuations in individual form could contain information that market probabilities do not fully reflect. This study therefore examined whether the recent recorded performances of identified starters improved three-way outcome predictions beyond a recalibrated closing market.

The analysis used 7,801 eligible top-division matches from Belgium, the Netherlands, Portugal, Scotland, and Turkey. A chronological development sample contained 6,638 matches from 2020/21–2024/25, while 1,163 matches from 2025/26 formed a final held-out test. Player and team histories were constructed from earlier matches. All feature choices, model settings, comparisons, and the decision not to apply post-hoc calibration were fixed before the final season was examined. The primary comparison placed a pooled market-only multinomial logistic model against an otherwise identical model augmented by four recent player-performance features. Each league received equal importance, and performance was assessed primarily through log loss, supported by Brier score, ranked probability score, country-level estimates, and paired match-week bootstrap intervals.

On the held-out season, the player-enhanced model recorded an equal-league log loss of 0.9580, compared with 0.9567 for the recalibrated market: a 0.143% deterioration. The paired 95% interval ranged from a 0.511% deterioration to a 0.234% improvement. Brier score and ranked probability score followed the same pattern, while country-level effects were small, mixed, and uncertain. These findings show no reliable incremental value from the tested player features and emphasize that player-enhanced forecasts require evaluation against strong market-only baselines on unseen data.

1 Introduction

1.1 Context and Significance

Probabilistic football forecasting is used to compare teams, support sporting decisions, and estimate the likelihood of home wins, draws, and away wins. It is also a demanding test of predictive modelling. Match outcomes are influenced by team quality, tactics, venue, scheduling, and the particular players available, while the relative importance of this information changes over time. A useful forecast must therefore do more than identify the most likely result: it must assign well-calibrated probabilities to all three possible outcomes.

Bookmaker closing odds provide a particularly strong benchmark for this task. They incorporate information from market participants and bookmakers until shortly before kick-off, including information about team strength, injuries, expected line-ups, and recent results. After removing the bookmaker margin, the resulting probabilities offer a compact summary of the information available to the market. Consequently, outperforming a simple historical baseline or a team-strength model is not sufficient evidence that a new input is valuable. The more demanding question is whether that input improves predictions beyond the closing market itself.

Player-level information is a plausible candidate for such an improvement. Football matches are played by changing groups of players, yet many forecasting models represent teams as stable units. Team-level histories may respond slowly to transfers, injuries, rotation, and changes in individual performance. Aggregating the recent recorded performances of the identified starting players could provide a more current description of the two teams entering a match. The practical value of this information, however, depends on whether it contributes something that closing odds do not already capture.

1.2 Research Gap

Existing approaches to football match prediction commonly use previous match results, goal-based team-strength estimates, or flexible statistical and machine-learning models. Player-level variables can be added to these systems, but an improvement over a team-only model does not establish that the player information is new to the market. When bookmaker probabilities are included, an apparent gain may also arise because the fitted model corrects systematic biases in those probabilities. A clean test must therefore compare a recalibrated market-only model with an otherwise identical model that additionally receives player information.

This question also requires careful separation between model development and evaluation. Player features, model settings, and calibration choices can appear effective when repeatedly assessed on the same seasons. Features calculated with information from after a match create a more direct form of bias. Both problems can produce improvements that would not have been available in a real forecasting setting. A credible assessment should construct every feature chronologically, make all modelling decisions using earlier seasons, and retain a final season that is examined only after the complete specification has been fixed. Testing several leagues under common rules further shows whether a result is broad or dependent on one competition.

1.3 Scope and Definition

We define Mid-Sized European Leagues as national first divisions in UEFA member associations that (i) are outside the Big Five markets (England, France, Germany, Italy, and Spain); (ii) belong to the intermediate tier of UEFA country coefficients—roughly ranks 6–20 over the last five seasons; and (iii) exhibit average club revenues in the middle clusters identified by UEFA and Deloitte (approximately €20–50 million per club per season), distinguishing them both from the Big Five and from low-revenue peripheral leagues.

The empirical study covers the top divisions of Belgium, the Netherlands, Portugal, Scotland, and Turkey. These competitions provide different scoring environments, draw rates, schedules, and numbers of fixtures while remaining within a comparable segment of European football. Applying common data and evaluation rules across them allows the analysis to distinguish an overall effect from results driven by a single league.

1.4 Research Question

The paper addresses the following question:

Does the recent recorded performance of identified starting players improve three-way football match predictions beyond recalibrated closing bookmaker probabilities?

The central test is deliberately incremental. Player information is considered useful only if a market-based model with player features performs better than the same recalibration model without them on identical matches. The final held-out evidence does not show a reliable improvement: the player-enhanced model is slightly worse overall, the country-level results are mixed, and the uncertainty intervals include no difference.

1.5 Approach and Contributions

The study constructs a common multi-league dataset under consistent inclusion rules for completed matches, closing odds, scores, starting line-ups, and player data. Player and team histories are updated chronologically within each league. The comparison framework ranges from historical outcome frequencies and Dixon–Coles team-strength models to raw closing probabilities, a trained market-only recalibration model, and several player-enhanced models. Shared models are trained across leagues while giving each league equal total importance; league-specific versions are fitted separately, and Dixon–Coles team strengths remain confined to their own competitions.

All feature and model decisions are made through chronological development experiments covering 2020/21–2024/25. The selected features, model settings, comparison rules, and decision not to apply post-hoc calibration are then fixed. The models are trained on those five seasons and evaluated once on 2025/26. Log loss is the primary measure, with Brier score and ranked probability score as secondary measures. Results are reported using equal-league, match-weighted, and individual-country summaries, while paired match-week resampling quantifies the uncertainty surrounding differences between models.

The paper makes three main contributions. First, it isolates the incremental value of player information through a direct comparison between otherwise identical market-based models. Second, it provides a chronological and auditable multi-league evaluation in which feature selection is separated from the final test. Third, it supplies evidence that recent recorded player-performance features, in their tested form, do not consistently improve on recalibrated closing bookmaker probabilities. The remainder of the paper reviews the related literature, describes the data and methods, presents the development and final results, and discusses their interpretation and limitations.

2 Related Literature

2.1 Football Match Prediction

Statistical football forecasting has traditionally represented a team through its previous results and scoring record. One foundational approach models the goals scored by each side as Poisson variables whose rates depend on the teams’ attacking and defensive strengths (Maher, 1982). This framework produces a full score distribution and, by aggregation, probabilities for a home win, draw, and away win. Dixon and Coles (1997) extended it by correcting the probabilities of low-scoring results and weighting recent observations more heavily. The resulting Dixon–Coles model remains an influential, interpretable benchmark that adapts as results are observed.

Later work developed richer representations of changing team strength. Rue and Salvesen (2000) treated team abilities as dynamic quantities in a Bayesian model, while Koopman and Lit (2015) allowed attacking and defensive strengths to evolve within a dynamic bivariate Poisson system. Other studies modelled the three-way outcome more directly. Goddard (2005), for example, compared regression models for goals and match results, and Hvattum and Arntzen (2010) used Elo ratings as inputs to match-result forecasts. Despite different assumptions, each summarizes history at team level.

Machine-learning research has widened the set of relationships that can be estimated from such data. Using English Premier League matches, Baboota and Kaur (2019) combined engineered performance features with several classification methods. Such models can capture nonlinearities and interactions, but their flexibility increases the choices made during development. More fundamentally, success against an outcome-frequency, Elo, or Poisson benchmark does not show that a model contains information absent from bookmaker prices. For the present research question, team-based models are therefore useful comparators, but the closing market is the more demanding reference point.

2.2 Bookmaker Markets as a Forecasting Benchmark

Fixed odds can be interpreted as probabilistic forecasts after accounting for the bookmaker’s margin. Their forecasting strength has been demonstrated in direct comparisons with statistical models. In almost 10,000 English matches, Forrest et al. (2005) found that a model containing a broad set of observable variables did not outperform the forecasts embedded in bookmaker odds. Štrumbelj and Robnik-Šikonja (2010) reached a similarly strong assess-

ment across six European leagues, while also showing that forecast quality differed between bookmakers and competitions and improved over time. These findings support the use of market probabilities as a benchmark without requiring the stronger claim that every quoted price is perfectly efficient.

Indeed, the literature contains evidence of both information efficiency and small, context-dependent departures from it. Dixon and Pope (2004) examined whether statistical forecasts retained value in the UK football betting market, whereas Angelini and De Angelis (2019) found differing degrees of efficiency across European online football markets. Market structure also matters: Franck et al. (2010) reported that betting-exchange prices were more accurate than the corresponding forecasts from bookmakers in their sample. Collectively, these studies suggest that odds are difficult to improve upon, but that their quality can depend on the provider, competition, and point at which they are observed. This motivates the use of the latest consistently available closing odds in the present study.

Raw inverse odds are not probabilities because their sum normally exceeds one. Removing this overround is therefore part of the forecasting problem rather than a purely cosmetic transformation. Štrumbelj (2014) showed that alternative methods for recovering probabilities from odds can produce different calibration and forecasting properties. A second distinction is equally important here. A trained model supplied with bookmaker probabilities may improve their log loss simply by learning systematic recalibration, even if every additional football feature is uninformative. Consequently, a market-plus-player model must be compared with a market-only model that receives the same opportunity to recalibrate the odds. Comparing only with raw probabilities would not identify whether an apparent gain came from player information or from recalibration.

2.3 Player-Level Information

Team-level histories treat the club as the main forecasting unit, even though its strength depends on the players selected. Player research has approached this problem from two broad directions. Bottom-up systems summarize recorded actions, while top-down systems infer a player’s underlying contribution from the results produced while that player is on the pitch. An early example of the first direction is the event-based player performance rating developed by McHale et al. (2012). In the second, plus-minus models distribute changes in team performance across the participating players while adjusting for teammates and opponents. Kharrat et al. (2020) developed goal-, expected-goal-, and expected-points-based versions of such ratings for football.

Starting line-ups provide a natural way to translate player estimates into team forecasts. Rather than estimating a separate fixed strength for every team, Maystre et al. (2016) related matches through the players appearing in their line-ups, allowing information to be shared across club and international competitions. Arntzen and Hvattum (2021) compared team Elo ratings with aggregated starting-player plus-minus ratings. The two sources performed similarly when used separately, but forecasts combining them were significantly better than forecasts based on either source alone. This result demonstrates that player and team ratings can contain complementary information, although it does not by itself establish improvement beyond the betting market.

The closest precedent for that stronger comparison is Holmes and McHale (2024), who

modelled changing player ability, aggregated the selected players into team strength, and evaluated match-result forecasts against bookmaker predictions. Their approach treats ability as a latent quantity that evolves through time. The present study asks a narrower but distinct question: whether recent directly recorded performance from the starting players adds value. That distinction matters because observed actions may be more transparent than a latent rating but also substantially noisier. Consistent with this concern, Gelade and Hvattum (2020) found that event-level statistics explained part of the variation in plus-minus ratings but produced only a marginal improvement in their match-prediction power.

Recent form also depends on how performance history is summarized. He et al. (2024) compared short rolling histories when forecasting individual match actions and showed that predictability varied with the action, playing position, and amount of recent information used. This supports treating window length as a modelling choice rather than assuming that more history is always better. It also highlights a limitation of rolling observed performance: such features mix persistent player quality with temporary form. The features tested in this paper deliberately represent recent recorded output; they are not presented as estimates of a player’s underlying ability.

2.4 Methodological Considerations

The literature suggests several requirements for isolating the contribution of player information. First, evaluation must respect time. Dynamic team and player ratings are meaningful only when every estimate is based on matches preceding the prediction. Randomly mixing earlier and later matches can give a model access to future team or player information, while even chronological splits become invalid if a rolling feature includes the match being predicted.

Second, feature and model decisions must be separated from the final assessment. Trying many feature groups, algorithms, and calibration choices on the same test period makes the reported result partly a product of that period. A sequence of chronological development folds can guide these decisions, but a final season must remain untouched until the specification is fixed. Third, models must be compared on identical matches. Differences in line-up or odds coverage can otherwise be mistaken for differences in forecast quality.

Finally, the comparison should reflect both the research question and the multi-league setting. A market-plus-player model isolates player value only when its market-only counterpart has the same structure and recalibration opportunity. Overall results should not be dominated by the league contributing the most fixtures, and country-level estimates should accompany the pooled result. Because small differences in probabilistic scores can arise from the particular sequence of matches observed, point estimates should also be paired with uncertainty intervals that preserve the grouping of fixtures in time.

2.5 Literature Gap and Positioning

Previous work establishes that team-strength models are useful, bookmaker odds are formidable forecasts, and player ratings can improve models built only from team information. It does not fully resolve whether short-term, recorded player performance adds information beyond

a recalibrated closing market. The closest studies generally estimate latent player ability, focus on major competitions, or answer whether player ratings complement team ratings rather than whether recent player output complements closing odds. This paper addresses that gap through a common five-league dataset, chronological feature construction, market-only and market-plus-player models with identical recalibration capacity, development decisions fixed before a held-out season, and paired uncertainty estimates on the final comparison.

3 Data

3.1 Countries and Seasons

The study covered the national top divisions of Belgium, the Netherlands, Portugal, Scotland, and Turkey over six seasons from 2020/21 to 2025/26. These competitions were processed under one data specification. The first five seasons served as the development and training period. The 2025/26 season was stored separately until the full modelling specification had been fixed.

Greece was collected and audited but excluded from the modelling sample before the comparative experiments. Its 2020/21 data contained no matches with complete starting line-ups, and only one match met that requirement in 2021/22. This made the common chronological development folds impossible. The exclusion followed the declared line-up rule rather than model performance. Regular-season, Scottish post-split, and applicable top-division play-off fixtures were retained when eligible.

3.2 Match and Odds Data

Match schedules, dates, teams, completion status, scores, line-ups, and player-match records were obtained from TheStatsAPI (TheStatsAPI, nd). Historical three-way odds were obtained from Football-Data.co.uk (Football-Data.co.uk, nd). Records were joined within league and season using match date and normalized team identities. A match entered the sample only when the final home and away scores agreed across both sources.

Only rows explicitly identified in the odds data as closing prices were used. All three decimal odds had to be present and greater than one. The bookmaker margin was removed by taking the inverse of each price and dividing it by the sum of the three inverse prices. This produced positive home, draw, and away probabilities that summed to one. They were treated solely as forecasts; the study did not simulate betting returns.

3.3 Player and Starting-Line-Up Data

The player records contained a stable player identifier, team assignment, position, starting status, minutes played, and event-level match statistics. These included goals, assists, shots, key passes, tackles, interceptions, saves, and provider ratings. Expected-goal fields were audited separately, but non-penalty expected goals were not used in the final active feature set. The final prediction sample required exactly eleven starters for the home team and eleven

for the away team. Every starter required a populated identifier and recorded minutes, and every player’s team identifier had to correspond to one of the two clubs in the fixture.

Player histories were formed independently inside each league. Under the primary rule, a player entering a new country contributed only appearances recorded in the current league; earlier performances from a foreign league were not carried across competitions. This avoided imposing an unestimated conversion between different league standards. Matches that lacked odds or otherwise failed as prediction targets could still contribute an earlier player appearance when a valid player record existed. Thus, excluding a match from forecast evaluation did not erase information available before a later match. Historical-variable construction is described in Section 4.

3.4 Identifiers and League Separation

Every combined record carried an explicit league field. League names were also prefixed to match and team identifiers before the country datasets were joined. This prevented unrelated clubs or source identifiers from colliding in the pooled table. Team-strength histories remained within leagues; pooling occurred only when fitting the specified shared models.

3.5 Inclusion Criteria and Construction

A match was eligible when it was a completed top-division fixture; appeared in both match sources; had identical final scores across them; contained valid closing odds for all three outcomes; contained player data; had eleven starters per side; had complete starter identities and minutes; and contained valid player-to-team assignments. These checks were applied match by match under a single rule set. Each excluded fixture retained a primary reason and its failed checks for audit.

Construction proceeded separately by league. After validating matches, odds, players, and line-ups, pre-match histories were created in date order with the current match excluded. The league tables then received league-safe identifiers, were combined, and were split into development and held-out files. Detailed feature calculations appear in the Methodology section.

Figure 1 shows how the common rules reduced the linked development sample to the 6,638 matches used for model development and final training. The figure refers to the first five seasons because 2025/26 was kept outside every development decision.

3.6 Data Coverage

The resulting sample contained 7,801 eligible matches. Of these, 6,638 belonged to the 2020/21–2024/25 development and training period, and 1,163 belonged to the final 2025/26 evaluation. Table 1 reports the retained matches by league and season. Coverage was generally stable, although Portugal contributed only 64 eligible matches in 2023/24 under the common rules. This shortfall was retained transparently rather than repaired through weaker country-specific requirements.

Figure 2 separates player-match availability, complete starting line-ups, and final model-ready coverage during development. It makes the Portuguese 2023/24 shortfall visible and

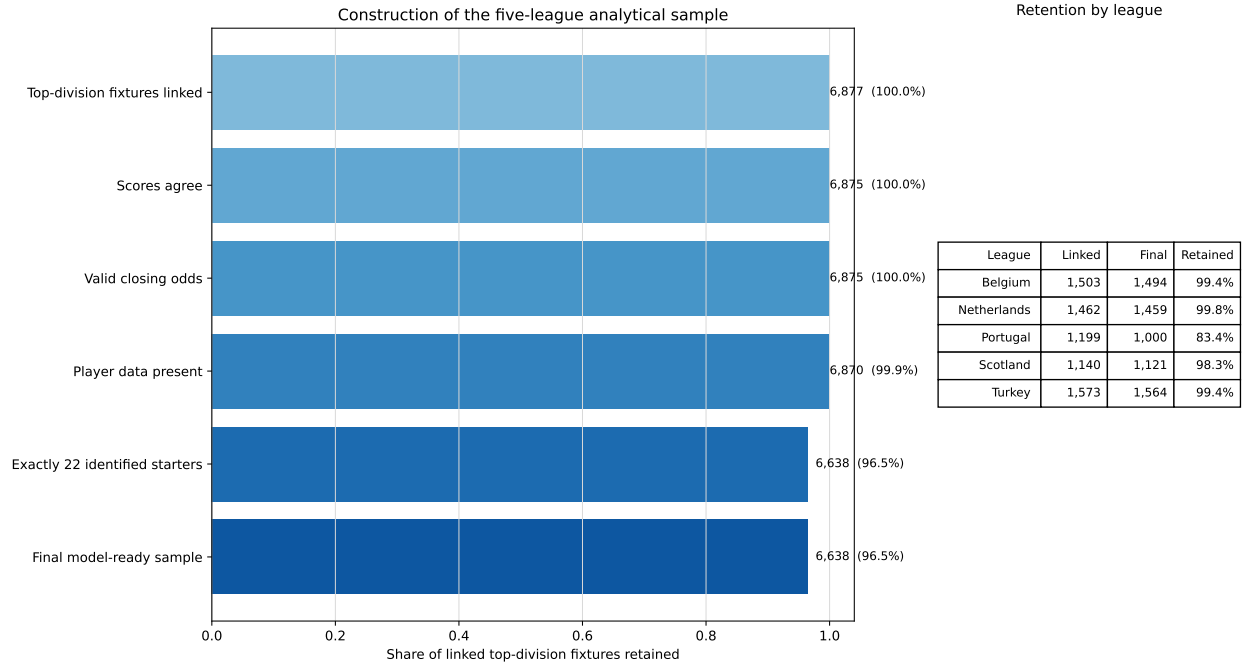


Figure 1: Construction of the five-league development sample. The same eligibility rules were subsequently applied to the held-out 2025/26 season.

Table 1: Eligible matches by league and season.

League	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	Total
Belgium	267	302	304	312	309	267	1,761
Netherlands	279	283	288	306	303	255	1,714
Portugal	238	239	240	64	219	211	1,211
Scotland	228	228	226	212	227	184	1,305
Turkey	341	338	277	339	269	246	1,810
Total	1,353	1,390	1,335	1,233	1,327	1,163	7,801

records that Greece was excluded under the predeclared line-up rule rather than after inspecting model results.

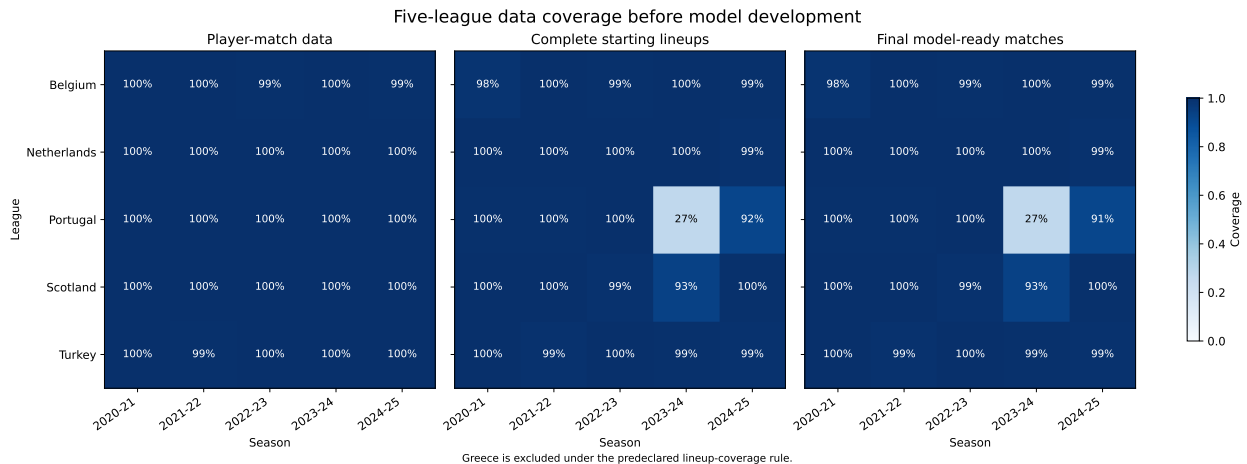


Figure 2: Development-period data coverage by league and season. Percentages are relative to linked top-division fixtures in each league-season.

The strict complete-case definition improved comparability between models because every forecast used the same matches and available inputs. It also reduced representativeness: excluded fixtures were more likely to come from league-seasons with incomplete line-ups, missing odds, or weaker player-data coverage. Coverage tables and exclusion counts are therefore reported alongside the modelling results rather than treating the retained sample as a complete record of every match played in the five competitions.

4 Methodology

4.1 Prediction Target

The forecasting target was the result of each match after normal time, expressed as one of three mutually exclusive outcomes: home win (H), draw (D), or away win (A). Every method produced a probability for each outcome, and the three probabilities were required to be positive and sum to one. The analysis was therefore concerned with the quality of the complete probability forecast rather than only whether the most likely outcome was correctly identified.

All predictors were defined from information available before the relevant match. Closing odds and announced starting line-ups represented the final pre-match information state. Match results were used only as targets or to update histories for later fixtures. Matches sharing the same kick-off time were processed as a group: their pre-match features were recorded before any result from that group was added to a player or team history.

4.2 Chronological Feature Construction

Player appearances were ordered by kick-off time within each league. For every starter, recent performance was calculated after shifting the appearance history by one match, so the current match could not contribute to its own predictors. The main rolling window contained the player’s previous five appearances, with shorter histories retained for new or infrequently observed players. A player’s record from another country was not carried into the new league. This rule avoided treating the same recorded performance as directly comparable across competitions with different playing environments and data coverage.

Shots, key passes, and defensive actions—defined as tackles plus interceptions—were converted to rates per 90 minutes before being summarized. For a given starting eleven, the previous-five-appearance rate of each starter was summed to obtain a line-up total. Separate home and away values were first constructed, after which the signed home-minus-away difference was used by the primary logistic models. Thus, for feature z , the match-level value was

$$z_m = z_{m,\text{home}} - z_{m,\text{away}}.$$

The feature-building system also generated alternative windows, exponentially weighted summaries, opponent-adjusted measures, positional summaries, line-up continuity measures, and indicators for limited player history. These variables formed the candidate pool during development; they were not all passed to the final models.

Team-strength variables were updated in the same chronological manner and separately within each league. Each team began with an Elo rating of 1,500. A 60-point home adjustment and a K factor of 20 were used, with the update scaled by the logarithm of the goal margin. Attacking and defensive goal rates began at 1.35 goals and were updated after each match using an exponential weight of 0.20 on the latest observation. A pre-match expected-goals-strength value then combined the team’s attacking rate with its opponent’s defensive rate; the home value also received a fixed multiplier of 1.10. This is a team-history control, not a player-performance measure.

The primary player-enhanced specification ultimately retained the four home-minus-away variables shown in Table 2. Three describe the recent recorded output of the announced starters, while the fourth controls for recent team scoring strength.

Table 2: Predictors added to the closing-market inputs in the primary player-enhanced model.

Predictor	Construction
Key passes	Sum of each starter’s key passes per 90 minutes over his previous five appearances
Shots	Sum of each starter’s shots per 90 minutes over his previous five appearances
Defensive actions	Sum of each starter’s tackles plus interceptions per 90 minutes over his previous five appearances
Expected-goals strength	Chronological team attacking rate combined with the opponent’s defensive rate

Figure 3 summarizes the information flow. The two primary models received the same closing probabilities and were separated only by the selected non-market inputs, while every history was calculated before the match being predicted.

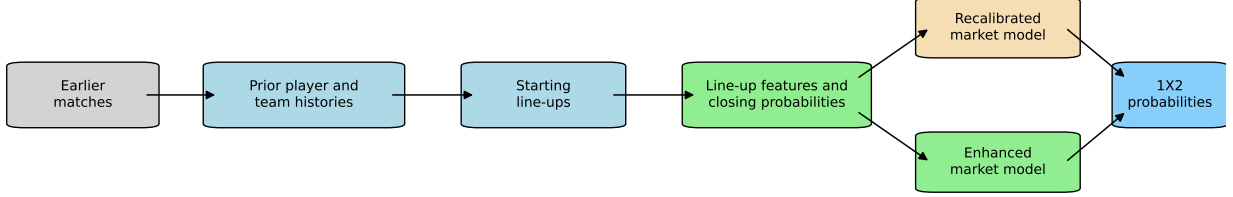


Figure 3: Chronological construction of inputs and the paired primary models.

4.3 Market Probabilities and Models

The inverse of each closing decimal odd was first calculated. The bookmaker margin was removed by dividing each inverse odd by the sum of the three inverse odds. If o_{mk} denotes the closing odd for outcome k in match m , the raw market probability was

$$p_{mk}^{\text{market}} = \frac{1/o_{mk}}{(1/o_{mH}) + (1/o_{mD}) + (1/o_{mA})}.$$

These normalized probabilities formed both a direct benchmark and the inputs to the learned market models. To avoid redundant information, the learned models used two log probability ratios, $\log(p_H/p_D)$ and $\log(p_A/p_D)$.

The primary comparison used two multinomial logistic models with the same estimation procedure. The recalibrated-market model received the two market log ratios and league indicators. It could learn systematic differences between the quoted probabilities and observed outcome frequencies, including league-specific scoring and draw environments. The player-enhanced model received those same inputs plus the four variables in Table 2. Comparing the two therefore tested whether the selected non-market information improved the same market recalibration model. Continuous inputs were standardized using the training sample. Coefficients were estimated with the multinomial logistic loss and L_2 regularization ($C = 1$), using the limited-memory BFGS solver with a maximum of 2,000 iterations.

Supporting models broadened the comparison without replacing this primary test. They included historical H/D/A frequencies; a logistic model using non-market features only; and two deterministic LightGBM classifiers, one based on home-minus-away variables and one retaining separate home and away values. The LightGBM models used 100 trees, a learning rate of 0.03, maximum depth three, seven leaves, a minimum of 30 observations per leaf, and an L_2 penalty of 1.0. Their random seed was fixed at 42.

The score-based benchmark followed Dixon and Coles (1997). Separate attacking and defensive parameters were estimated for each team, together with home advantage and a correction for low-scoring outcomes. Older observations received exponentially declining weight with a half-life of 365 days. Score probabilities from zero to ten goals for each side

were calculated and then combined into H/D/A probabilities. A player-enhanced version added a regularized linear shift based on selected home-minus-away player variables to the two teams’ expected goal rates. Dixon–Coles models were always estimated separately by league because attack and defence parameters have meaning only within the competition in which the teams played.

4.4 Pooled and League-Specific Training

For models that could share information across countries, the primary training scope pooled all five leagues. League indicators allowed their baseline outcome patterns to differ, while the remaining coefficients were shared. Because the number of eligible fixtures differed by competition, matches were weighted so that every league contributed the same total training weight. If N is the number of training matches, $L = 5$ is the number of leagues, and n_l is the number from league l , each match in that league received weight $N/(Ln_l)$. A larger league could therefore not dominate estimation merely by providing more fixtures.

The same model families, input definitions, and fixed settings were also fitted separately in Belgium, the Netherlands, Portugal, Scotland, and Turkey. These league-specific estimates served as a secondary analysis of whether sharing information was beneficial and whether performance differed across countries. They did not affect the specification selected for the primary pooled test.

4.5 Development Procedure

All modelling decisions were made using the first five seasons. Three expanding-window folds reproduced the information that would have been available at the time of prediction, as shown in Table 3. No later development season was used to fit a model predicting an earlier one.

Table 3: Chronological development folds.

Prediction season	Training seasons
2022/23	2020/21–2021/22
2023/24	2020/21–2022/23
2024/25	2020/21–2023/24

Feature selection was conducted separately for each model family. Candidate variables were first organized into interpretable groups, and all valid group combinations were evaluated through the same three folds. Mean equal-league log loss was the selection criterion. When several combinations were within 0.005 log-loss units of the best result, the smaller set was preferred. The two strongest starting sets were then refined through at most six accepted-search steps. At each step, every permissible one-feature addition and removal was evaluated, and the best move was accepted only if it reduced mean log loss by more than 0.0005. Major feature-block removals, a within-league-and-season player shuffle, and a deliberate future-information test were subsequently used to check that apparent gains were not dependent on one broad input class, arbitrary player assignment, or leakage. The

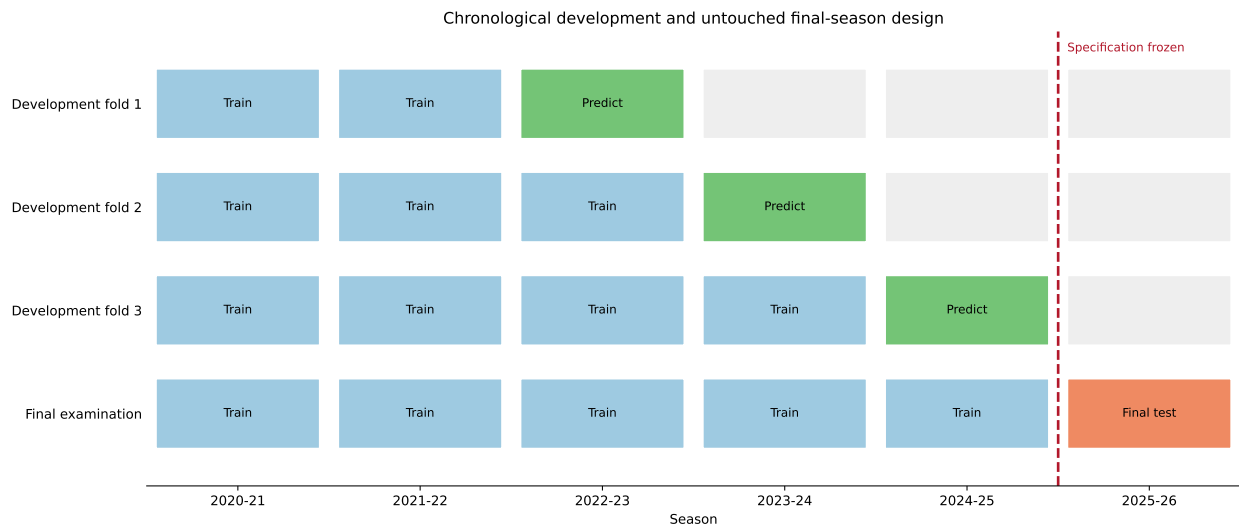


Figure 4: Expanding-window development and the untouched final-season evaluation. The vertical boundary marks the point at which the specification was fixed.

final block-removal check led to removal of the five-appearance mean player rating from the primary market-plus-player model.

The selected feature lists were saved and loaded by both pooled and league-specific evaluators rather than being reselected during later runs. Model settings, feature counts, and a checksum of the saved feature file were also recorded. Tests rejected missing, duplicated, or altered feature entries. Post-hoc probability calibration was examined on development predictions but was not retained because it did not provide a consistent improvement. The final procedure therefore applied no temperature scaling or other adjustment after model fitting.

Figure 5 shows the complete group-search results in compact form. Each blue point is an evaluated group combination and the red line traces the best combination available at each level of complexity. The different vertical scales are deliberate: they show variation relative to the best result inside each model family rather than suggesting that the families had equal absolute performance. Full combination maps are reported in Appendix D.

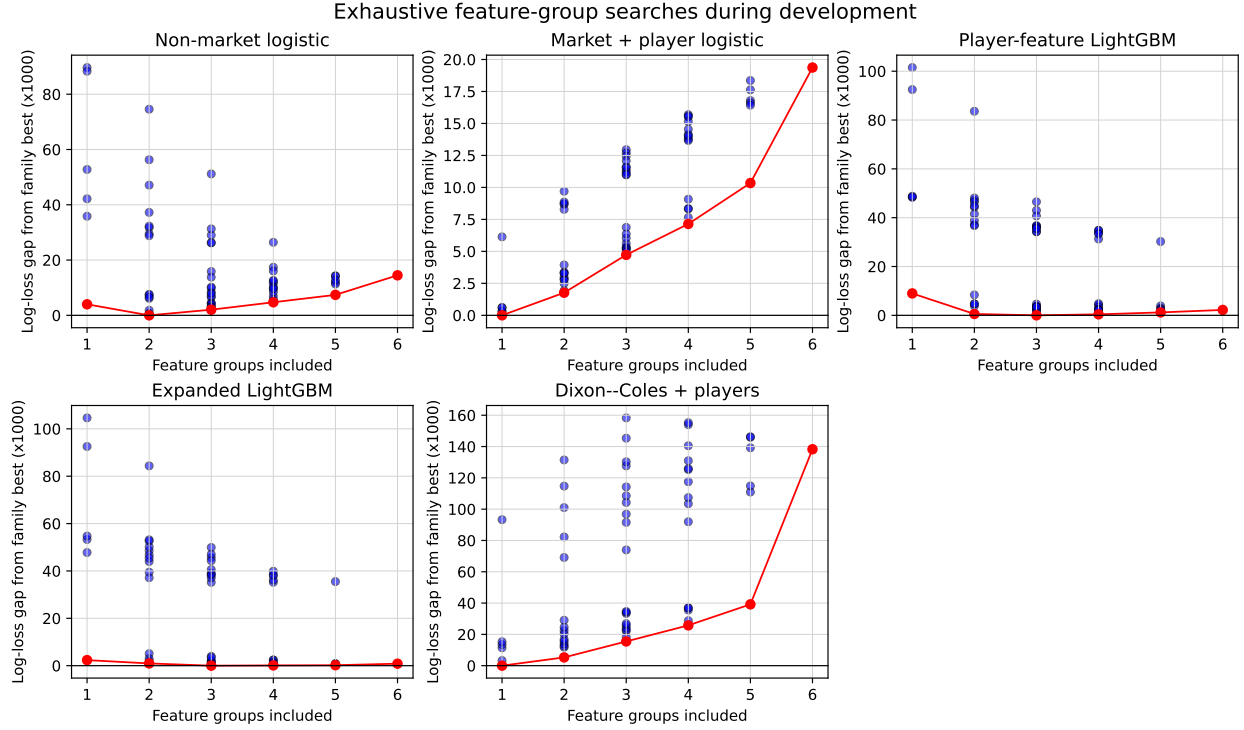


Figure 5: Exhaustive feature-group searches across the five model families. Vertical values are the log-loss gap from the best candidate within the same family, multiplied by 1,000.

4.6 Final Evaluation

After development was complete, the feature lists, model settings, training scopes, comparison rules, scoring measures, league weighting, and uncertainty procedure were fixed in a separate final-evaluation configuration. Each final model was trained once on all eligible matches from 2020/21–2024/25 and then used to predict the previously held-out 2025/26 season. The primary pooled model continued to use equal-league training weights, whereas Dixon–Coles remained league-specific.

Model coefficients were not updated in response to 2025/26 outcomes. However, the chronological player and team histories for a later 2025/26 fixture could naturally include information from earlier fixtures in that season, matching a real forecasting process. Automated checks prevented overlap between the training and final seasons, verified the saved feature checksum, required every model to cover the same eligible matches, and refused to overwrite an existing final-output directory. The final season was not used to revise features, settings, or calibration after its results were observed.

5 Comparison Framework

The comparison framework was organized as a ladder. Each step introduced a stronger source of information or a more flexible way of using it. This design made it possible to distinguish several questions that would otherwise be combined: whether statistical modelling improved

on a simple historical guess, whether player information improved a conventional team-strength model, whether any model approached the closing market, and whether selected non-market features added information after the market itself had been recalibrated. All methods predicted the same eligible matches, so differences in performance could not be attributed to differences in sample coverage.

Figure 6 displays the sequence and the interpretation of each comparison. The rungs are not assumed to improve monotonically; they separate the information source being tested from the flexibility of the model using it.

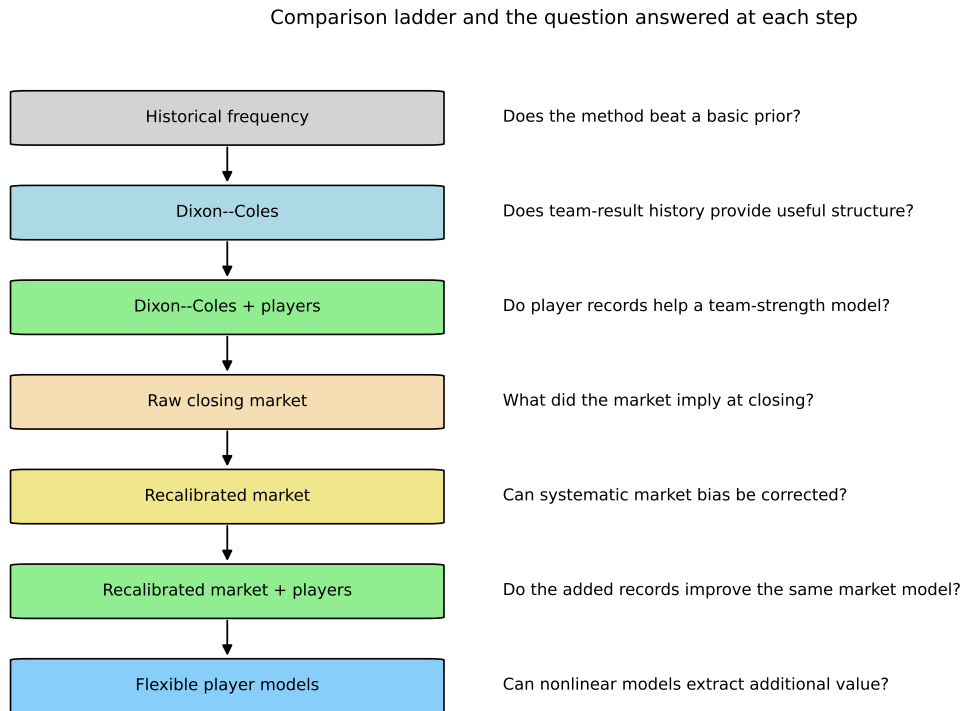


Figure 6: Comparison ladder used to identify where any forecasting improvement originated.

5.1 Simple Baselines

The historical-frequency model formed the lowest rung of the ladder. It assigned probabilities according to the home-win, draw, and away-win frequencies observed in the training data. In pooled evaluation, these frequencies were maintained separately for each league; in league-specific evaluation, they were calculated from that league’s training matches alone. The model used no information about the teams, players, current fixture, or bookmaker prices.

This baseline was not intended to be competitive with the closing market. Its purpose was to verify that the more elaborate methods extracted useful match-specific information rather than merely reproducing the average outcome distribution. A method that could not improve on historical frequencies would have little value as a forecasting model, regardless of its complexity.

5.2 Team-Strength Models

Plain Dixon–Coles provided the principal non-market team-strength benchmark. It used earlier scores to estimate each team’s attacking and defensive strength, home advantage, and the dependence between low-scoring outcomes. It therefore represented what could be predicted from the historical performance of the two teams without access to player data or bookmaker probabilities.

The corresponding player-enhanced Dixon–Coles model added selected summaries of the announced starting line-ups. Comparing it directly with plain Dixon–Coles tested whether player information improved a conventional score-based forecasting system:

Dixon–Coles with players versus plain Dixon–Coles.

Both versions were fitted separately within each league. Their role was supporting rather than primary because neither incorporated closing odds. An improvement over plain Dixon–Coles would show that player information helped a team-strength model, but it would not establish that the same information was absent from bookmaker prices.

5.3 Market Models

The first market benchmark consisted of the raw closing probabilities after the bookmaker margin had been removed. These probabilities required no fitting and represented the market’s forecast shortly before kick-off. They provided a much more demanding reference than either historical frequencies or a model based only on previous scores.

The second market benchmark was the recalibrated-market model. It received the closing home, draw, and away probabilities, expressed through two log probability ratios, together with league indicators in pooled training. It was allowed to learn recurring differences between the quoted probabilities and the observed outcomes. For example, it could adjust a systematic tendency to assign too much or too little probability to favourites or draws.

Including both market benchmarks was necessary for interpreting any apparent gain from a fitted player model. A trained model could outperform the raw closing probabilities simply because it corrected a small market calibration bias. Such a result would not demonstrate that player information was useful. The recalibrated market therefore served as the primary benchmark against which the incremental non-market variables were assessed.

5.4 Player-Enhanced Models

Three forms of player-enhanced prediction completed the ladder. First, a multinomial logistic model used selected player and team-strength variables without bookmaker probabilities. It showed how much these variables could predict independently, but was not a direct test of information beyond the market.

Second, the primary player-enhanced model used the same multinomial logistic structure as the recalibrated-market benchmark and added the four selected non-market predictors. Three of these summarized the recent key passes, shots, and defensive actions of the announced starters. The fourth was the chronological expected-goals-strength difference derived from team results. The central comparison was therefore

recalibrated market + selected non-market predictors versus recalibrated market only.

Because the two models otherwise used the same training data, market inputs, league controls, estimation procedure, and weighting, their difference measured the incremental contribution of the added predictor set rather than a general difference in modelling method. The interpretation nevertheless requires an important qualification: the added set contained three player-performance variables and one team-strength control. The primary contrast consequently measured their combined contribution. The feature-block robustness analysis, which removed the team-strength variable while retaining the player variables, provided the more specific check of whether recent recorded player performance alone accounted for any improvement.

Finally, two LightGBM models examined whether nonlinear relationships or interactions could extract information that the logistic models missed. One used home-minus-away feature differences, while the other retained separate home and away values. These models were treated as secondary comparisons. Their greater flexibility could reveal additional predictive structure, but also made it harder to attribute an improvement to a particular source of information.

Taken together, the ladder addressed six increasingly demanding questions: whether the models beat historical outcome frequencies; whether player information improved Dixon–Coles; whether statistical models approached the raw closing market; whether recalibration improved the raw probabilities; whether selected non-market predictors improved an otherwise identical recalibrated-market model; and whether a more flexible learner altered that conclusion. The fifth comparison was the main test, while the remaining rungs provided context and checks on its interpretation.

6 Evaluation

6.1 Scoring Measures

Forecasts were evaluated as probabilities rather than only by whether their most likely outcome was correct. Lower values indicated better performance for all three scoring measures. Log loss was the primary measure because it assessed the probability assigned to the outcome that occurred and imposed a substantial penalty on confident incorrect forecasts. For N matches, it was calculated as

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N \log(p_{i,y_i}),$$

where p_{i,y_i} is the probability assigned by the model to the observed outcome of match i .

The multiclass Brier score was the first secondary measure. If p_{ik} is the predicted probability of outcome k and y_{ik} equals one when that outcome occurred and zero otherwise, the score was

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N \sum_{k \in \{H, D, A\}} (p_{ik} - y_{ik})^2.$$

Unlike log loss, the Brier score penalized probability errors quadratically and was less strongly affected by an individual confident mistake. The ranked probability score (RPS) was also reported. It compared the cumulative predicted and observed probabilities across the ordered outcomes H, D, and A, normalized by the two cumulative boundaries:

$$\text{RPS} = \frac{1}{N} \sum_{i=1}^N \frac{1}{2} \sum_{j=1}^2 \left(\sum_{k=1}^j p_{ik} - \sum_{k=1}^j y_{ik} \right)^2.$$

Agreement across log loss, Brier score, and RPS reduced the risk that the conclusion depended on one particular penalty for probability errors. Accuracy was used only as a descriptive development diagnostic because it ignored the size and quality of the predicted probabilities. It did not determine the final conclusion.

6.2 League Weighting

Performance was summarized in three ways. The primary equal-league result was constructed by calculating a score separately within Belgium, the Netherlands, Portugal, Scotland, and Turkey and then taking the unweighted mean of the five league scores. Each country therefore contributed one fifth of the result, regardless of its number of eligible fixtures. This aggregation matched the multi-league research question and prevented the largest sample from determining the overall conclusion.

The match-weighted result pooled all predictions before calculating each score. It described performance for the average match in the retained sample but gave greater influence to leagues with more fixtures. Results were also reported for each league individually to reveal whether an overall average concealed opposing country-level effects. These two views were secondary to the equal-league result.

Evaluation weighting was distinct from training weighting. During pooled model fitting, each league was assigned equal total weight so that it contributed equally to coefficient estimation. During evaluation, the equal-league summary gave the five resulting league scores equal importance. League-specific models were trained only on their own competition but were assessed through the same three reporting views wherever a combined summary was required.

6.3 Uncertainty Intervals

Point estimates alone could not show whether a small difference between two models represented a stable advantage or ordinary variation between match weeks. Uncertainty was therefore measured through a paired match-week bootstrap. The baseline and enhanced models were first matched by league, season, fixture, date, and observed outcome. Any comparison with missing or additional matches on either side was rejected.

Calendar weeks were then sampled with replacement within every league-season. All matches from a selected week were retained together, and the same sampled matches were used for both models. Resampling weeks rather than individual fixtures preserved some of the common conditions affecting matches played at a similar time, while pairing prevented differences in fixture difficulty from being mistaken for differences between models. The procedure used 10,000 repetitions and random seed 42.

For each repetition and score, improvement was defined as

$$\text{improvement} = \text{baseline score} - \text{enhanced-model score}.$$

A positive value therefore favoured the enhanced model, while a negative value favoured the baseline. Absolute and percentage improvements were retained. The 2.5th and 97.5th percentiles of the bootstrap distribution formed the reported 95% interval. An interval entirely above zero supported an improvement; an interval entirely below zero supported deterioration; and an interval containing zero indicated that the available matches did not clearly distinguish the two models. Equal-league and match-weighted intervals were calculated, while individual-country intervals were reported as supporting evidence and were not used for further model selection.

The principal interval compared the pooled recalibrated-market model with the pooled market-plus-selected-features model. The same paired procedure was used for secondary contrasts, including recalibration versus the raw closing market. Intervals quantified sampling uncertainty in the observed fixtures; they did not account for every possible alternative data source, feature definition, or modelling decision.

6.4 Robustness Checks

Four focused checks were conducted during development before the final specification was applied to 2025/26. First, the primary market-only and market-plus-selected-features models were trained through the same folds and evaluated on identical matches. This ensured that their difference could not be explained by unequal samples, market inputs, training weights, or estimation procedures.

Second, the selected predictors were divided into attacking output, defensive output, and team-strength context. Each block was removed in turn and the reduced model was compared with the complete specification. Removing the team-strength block was particularly important because it left the recent key passes, shots, and defensive actions of the starters while excluding the expected-goals-strength control. This provided a more specific assessment of the player-performance variables than the four-feature primary bundle alone.

Third, the three player-performance variables were shuffled jointly between matches within each league and season, using seed 42. Joint shuffling preserved their internal relationship and approximate league-season distribution while breaking the link between the recorded starting players and the fixture they actually played. If the shuffled version performed similarly to the genuine ordering, this would weaken the claim that the identities and recent records of the actual starters supplied useful information.

Fourth, a deliberately invalid hindsight variable was constructed from each team’s final points per match over the complete season. Because this quantity used results that were

unavailable at the time of earlier fixtures, it was excluded from every legitimate model. Its sole purpose was to demonstrate how future information could create an artificial forecasting improvement and to check that the valid chronological pipeline did not admit such information.

Single-temperature post-hoc calibration was additionally tested using only earlier out-of-sample development predictions. The same chronological procedure was applied to both primary models. Calibration was not retained because the predeclared conditions for a consistent improvement were not met; the final evaluation therefore used the original model probabilities.

Taken together, the evaluation required more than a favourable point estimate. The main evidence was the change in equal-league log loss between the otherwise identical recalibrated-market and enhanced models. Brier score, RPS, match-weighted and country-level results, paired uncertainty intervals, and the robustness checks were used to determine whether its direction was consistent and whether any apparent advantage could reasonably be attributed to the selected player information.

7 Results

7.1 Development Results

The three chronological development folds produced 3,895 out-of-sample predictions across the five leagues. Table 4 compares the pooled primary models in each prediction season. The enhanced model recorded lower equal-league log loss than the recalibrated market in 2022/23 and 2023/24, but higher log loss in 2024/25. Across all three folds, its log loss was 0.9239, compared with 0.9248 for the recalibrated market and 0.9267 for the raw closing probabilities. The overall development advantage over recalibration was therefore 0.0009 log-loss units, or approximately 0.10%.

Table 4: Equal-league log loss for the pooled primary models during development. Lower values are better.

Prediction season	Raw market	Recalibrated market	Enhanced model
2022/23	0.9164	0.9151	0.9130
2023/24	0.9163	0.9107	0.9081
2024/25	0.9407	0.9403	0.9422
All folds	0.9267	0.9248	0.9239

Figure 7 makes the change in direction across seasons explicit. The enhanced model improved on recalibration in the first two folds but not in 2024/25, which is why the small pooled development advantage was not treated as settled evidence.

Pooling was beneficial during development for the enhanced specification. Its equal-league log loss was 0.9239 when one model was shared across leagues and 0.9322 when a separate model was fitted in each league. The direction was different for market-only recalibration: the league-specific version recorded 0.9263, compared with 0.9248 for the pooled version. The pooled specification remained the predeclared primary model.

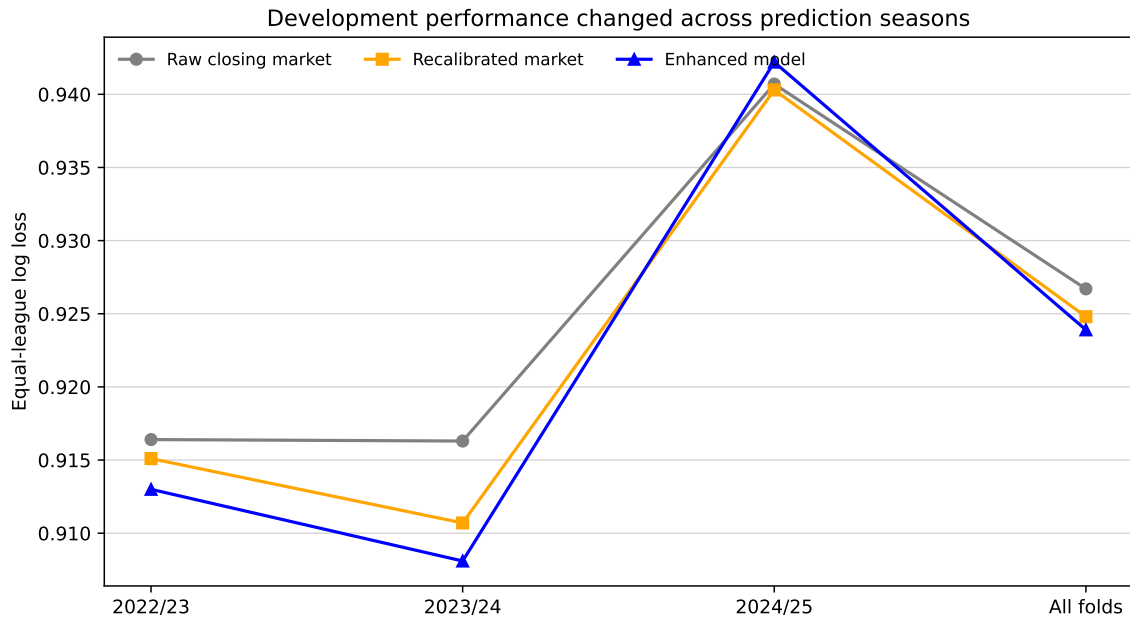


Figure 7: Equal-league log loss across the three chronological prediction seasons and over all development predictions.

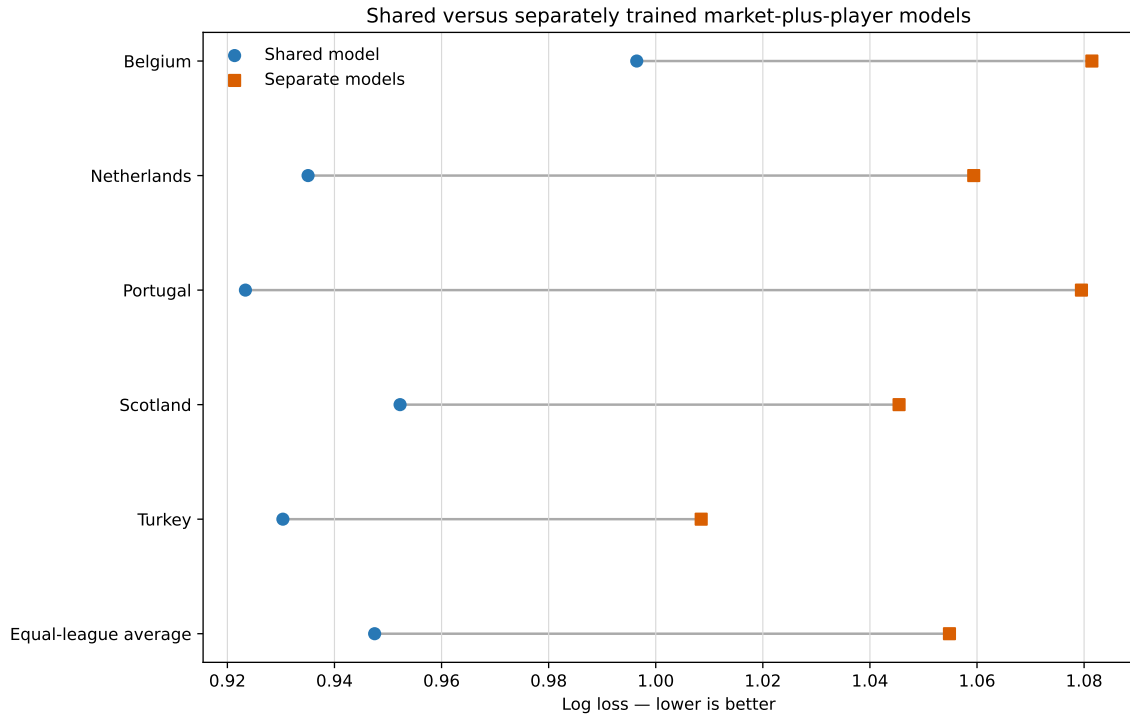


Figure 8: Development-period log loss for the shared and separately trained enhanced market models. Lower values are better.

The Dixon–Coles comparisons provided little development evidence of a player effect. The player-enhanced version achieved equal-league log loss of 0.9619, only 0.0002 below the 0.9621 of plain Dixon–Coles. Both were materially worse than the raw and recalibrated closing-market benchmarks. These comparisons were retained as supporting evidence rather than used to select the primary model.

7.2 Final Results

The fixed models were next trained on all 6,638 eligible matches from 2020/21–2024/25 and evaluated on the 1,163 previously held-out matches from 2025/26. Table 5 presents the primary pooled comparison. The enhanced model recorded an equal-league log loss of 0.9580, compared with 0.9567 for the recalibrated market. Because improvement was defined as the baseline score minus the enhanced-model score, the difference was -0.00136 , equivalent to a 0.143% deterioration. Brier score and RPS gave the same direction, with relative deteriorations of 0.210% and 0.254%, respectively.

Table 5: Primary pooled comparison on the held-out 2025/26 season. Relative change is positive when the enhanced model is better.

Measure	Recalibrated market	Enhanced model	Relative change
Log loss	0.9567	0.9580	-0.143%
Brier score	0.5697	0.5709	-0.210%
RPS	0.1884	0.1889	-0.254%

The paired match-week interval for the relative log-loss improvement extended from -0.511% to 0.234% . The interval included zero, and 22.9% of the 10,000 resamples favoured the enhanced model. The Brier-score interval ranged from -0.633% to 0.215% , while the RPS interval ranged from -0.828% to 0.327% ; both also included zero. The match-weighted estimates were similar. Under that aggregation, the enhanced model deteriorated by 0.154% in log loss, 0.233% in Brier score, and 0.277% in RPS, and none of the three intervals excluded zero.

Figure 9 shows the point estimates and paired match-week intervals together. All six estimates were negative and every interval crossed zero.

Table 6 places the primary result within the wider model suite. Recalibration improved on the raw closing probabilities in both pooled and league-specific training. The pooled non-market logistic and LightGBM models did not match the market-based models. Plain Dixon–Coles remained behind the market, and adding its selected player features increased rather than reduced held-out log loss. Historical outcome frequencies produced the highest log loss of the reported comparators.

The league-specific enhanced market model had the lowest descriptive log loss in the full table, 0.9542, compared with 0.9561 for league-specific recalibration. Its Brier score was also lower, at 0.5681 against 0.5688, whereas its RPS was fractionally higher, at 0.1880 against 0.1880 before rounding to four decimal places. This secondary result arose from five separately estimated models and did not replace the predeclared pooled comparison.

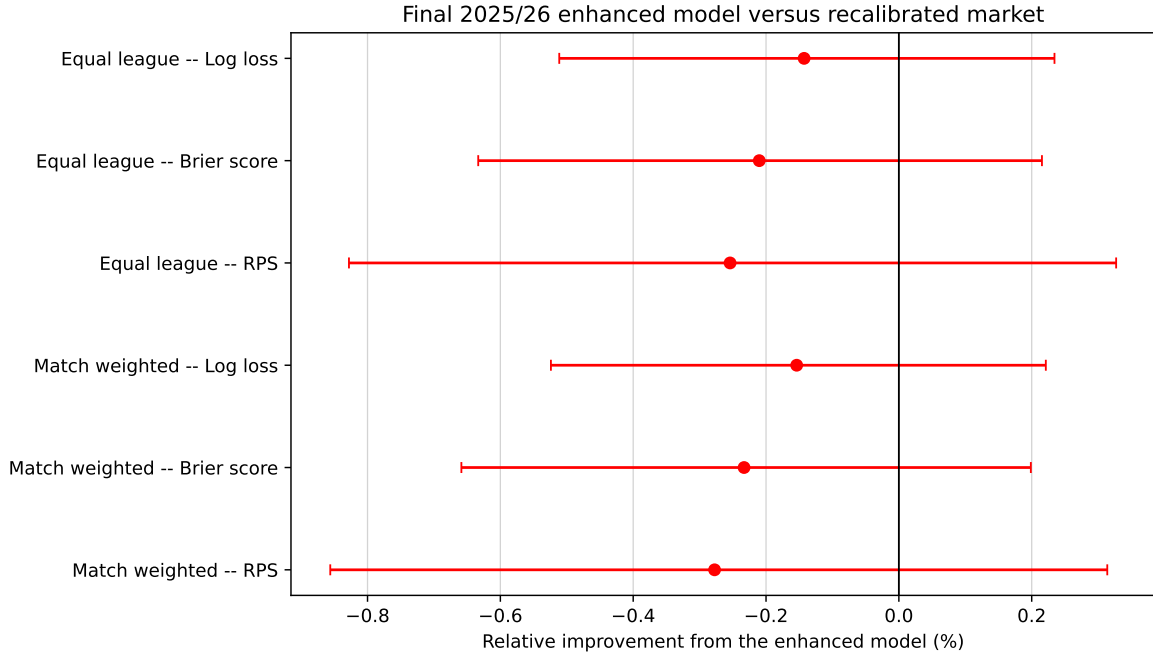


Figure 9: Relative improvement of the pooled enhanced model over the pooled recalibrated market in 2025/26. Positive values favour the enhanced model; bars are paired 95% match-week bootstrap intervals.

7.3 Country-Level Results

The pooled primary comparison varied in direction across countries (Table 7). The enhanced model reduced log loss in the Netherlands and Portugal but increased it in Belgium, Scotland, and Turkey. The largest point improvement occurred in Portugal, at 0.450%, while the largest deterioration occurred in Belgium, at 0.629%.

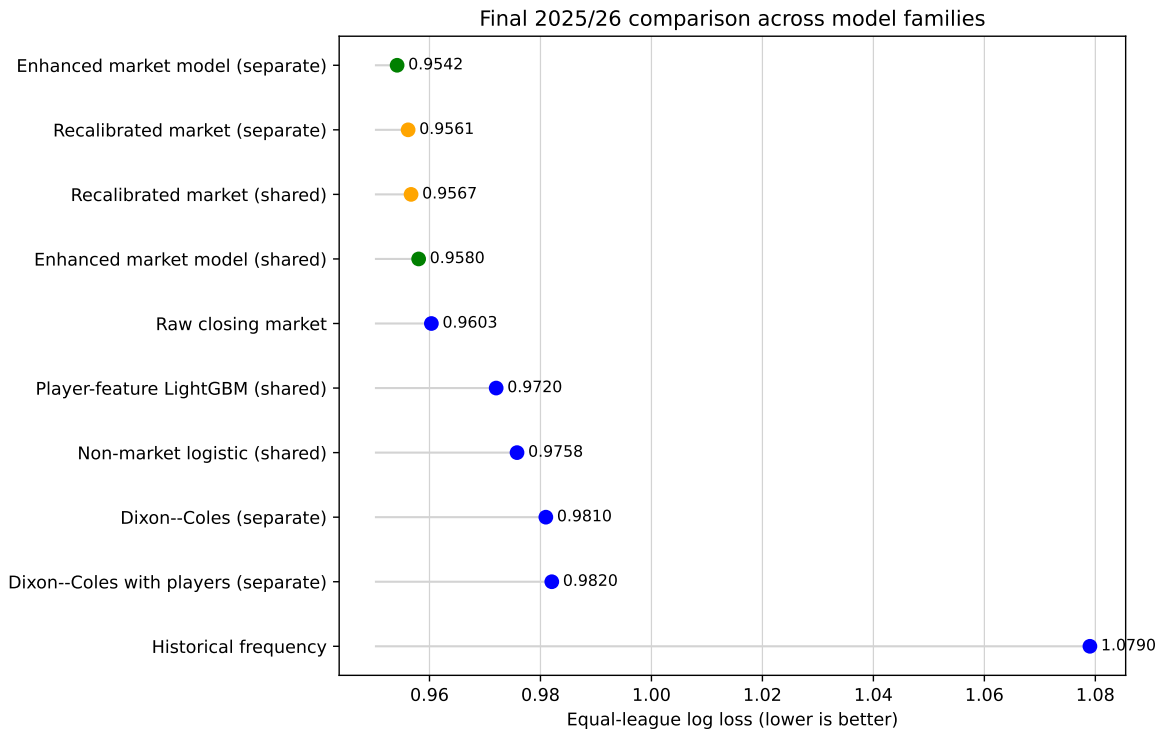


Figure 10: Equal-league log loss across the principal final comparators. Green marks denote enhanced market models, orange marks denote recalibrated-market models, and blue marks denote the remaining benchmarks.

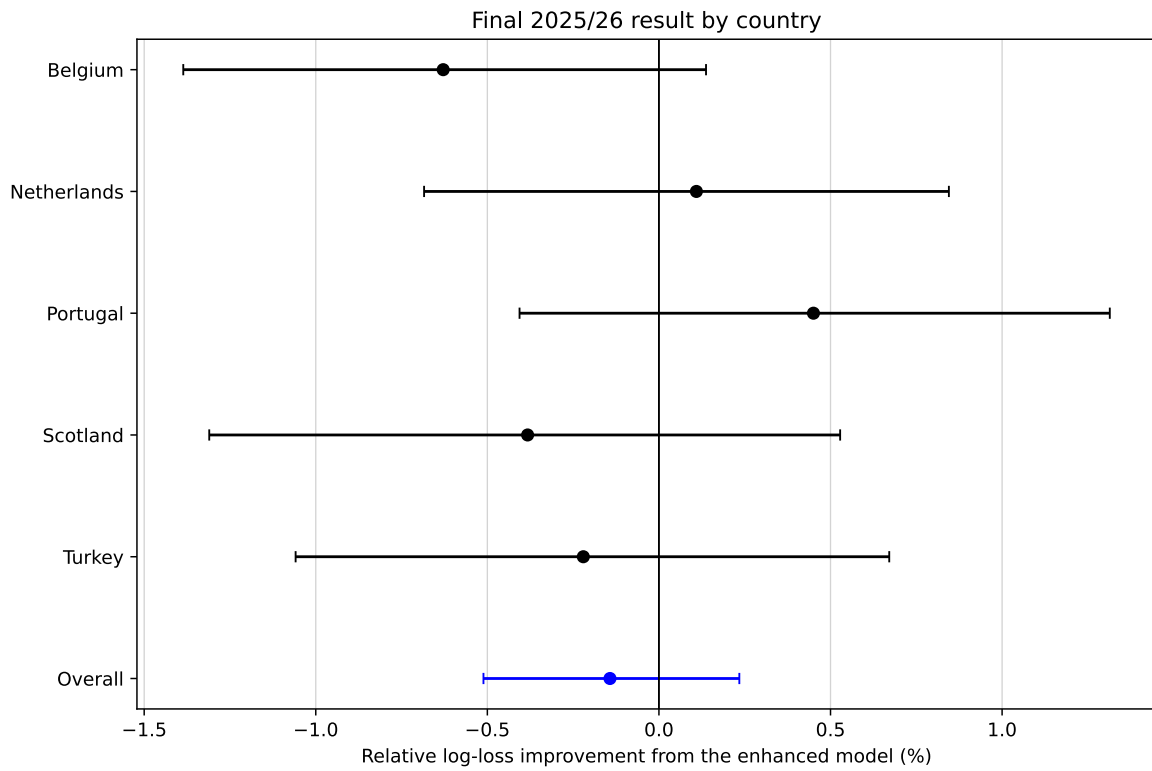


Figure 11: Country-level and equal-league relative log-loss changes for the pooled enhanced model against the pooled recalibrated market. Positive values favour the enhanced model.

Table 6: Equal-league log loss of the principal comparators in 2025/26.

Training scope	Model	Log loss
League-specific	Enhanced market model	0.9542
League-specific	Recalibrated market	0.9561
Pooled	Recalibrated market	0.9567
Pooled	Enhanced market model	0.9580
Pooled/league-specific	Raw closing market	0.9603
Pooled	Player-feature LightGBM	0.9720
Pooled	Non-market logistic model	0.9758
League-specific	Dixon–Coles	0.9810
League-specific	Dixon–Coles with players	0.9820
Pooled/league-specific	Historical frequency	1.0790

Table 7: Pooled primary log-loss comparison by league in 2025/26. Positive relative change favours the enhanced model.

League	Market	Enhanced	Relative change	95% interval
Belgium	0.9930	0.9992	−0.629%	−1.386%, 0.137%
Netherlands	0.9843	0.9833	0.109%	−0.684%, 0.845%
Portugal	0.9074	0.9033	0.450%	−0.406%, 1.314%
Scotland	0.9596	0.9633	−0.383%	−1.310%, 0.528%
Turkey	0.9391	0.9412	−0.220%	−1.059%, 0.671%

Every country-level interval included zero. The proportion of resamples favouring the enhanced model ranged from 5.5% in Belgium to 84.5% in Portugal, with values of 60.8% in the Netherlands, 20.8% in Scotland, and 31.1% in Turkey. The country results therefore showed mixed directions without an interval that clearly separated the two models.

7.4 Robustness Results

The focused robustness checks were conducted on development predictions before the held-out season was opened. The initial market-plus-player candidate contained five added variables, including the mean recent player rating. Removing that rating reduced overall log loss from 0.9241 to 0.9239 and improved each of the three development folds: from 0.9134 to 0.9130 in 2022/23, from 0.9084 to 0.9081 in 2023/24, and from 0.9423 to 0.9422 in 2024/25. It was therefore removed, producing the four-feature specification used in the final evaluation.

The remaining block checks were performed on the five-feature precursor. Removing the two attacking-output variables increased its log loss by 0.00002; removing the defensive-output variable increased it by 0.00057; and removing the team-strength variable increased it by 0.00035. These changes were small. Jointly shuffling the player variables within league and season increased log loss from 0.9241 to 0.9250. The correctly assigned player values therefore performed better than the shuffled version in development, although the difference was less than one thousandth of a log-loss unit.

The deliberately invalid hindsight model achieved log loss of 0.8928, compared with

0.9241 for the legitimate five-feature precursor. This large artificial gain arose from using each team’s final full-season points per match and was excluded from every valid comparison. Finally, chronological temperature scaling reduced neither primary model’s development loss consistently. It worsened equal-league log loss by 0.00158 for the recalibrated market and by 0.00210 for the enhanced model, so no post-hoc calibration was applied in the final evaluation.

In summary, development results showed a small aggregate advantage for the selected enhanced specification, but the advantage changed direction in the last development fold. It did not persist in the predeclared pooled evaluation of 2025/26: the enhanced model was slightly worse under all three scoring measures, country-level effects were mixed, and all relevant 95% intervals included zero.

8 Discussion

8.1 Interpretation

The study found no reliable evidence that the tested recent player-performance variables improved pooled match forecasts beyond recalibrated closing bookmaker probabilities. The selected enhanced model had a small advantage across the development folds, but that advantage was not consistent over time and reversed in the untouched 2025/26 season. On the final sample, log loss, Brier score, and RPS all favoured the market-only model by a small amount. The corresponding uncertainty intervals included both an improvement and a deterioration, while the country-level estimates pointed in different directions. The evidence therefore does not support a dependable incremental forecasting gain.

This conclusion is narrower than the claim that player information has no value in football prediction. The experiment evaluated a particular representation: recent recorded key passes, shots, and defensive actions from the announced starters, aggregated over each player’s previous five appearances. It shows that these summaries, used in the tested models and sample, did not add enough stable information to improve the primary market-based forecast. Different player data, longer histories, measures of underlying ability, tactical variables, or models of interaction between players may produce different results.

One likely explanation is that closing odds already reflected much of the information represented by the added variables. Prices observed shortly before kick-off can incorporate injuries, transfers, expected and confirmed line-ups, recent form, team news, and the beliefs of many market participants. The player-performance features may therefore have described genuine differences between the teams without describing differences that were new to the market. This distinction explains why a variable can be related to match outcomes yet still fail to improve a forecast conditional on closing probabilities.

A second explanation concerns measurement. Five-appearance event rates combine temporary form with match context. A player’s recent shots or key passes may change because of opponent strength, tactical instructions, playing position, score state, or minutes played rather than because the player’s ability has changed. Per-90 conversion and chronological construction reduce some problems, but they do not make recent event counts a pure measure of player quality. The resulting signal may be too noisy to correct an already strong

market forecast reliably.

Aggregation may also discard relevant structure. Summing the recent output of eleven starters treats the line-up mainly as a collection of individual totals. It does not fully represent whether the players suit particular roles, whether their strengths complement one another, or how a manager is likely to use them. The same number of shots has a different meaning for a forward and a full-back, and a line-up’s value may depend on partnerships that cannot be recovered from simple sums. The wider candidate set contained positional and continuity measures, but the final selection did not establish a stable gain from that additional complexity.

The difference between development and final performance is itself informative. The development advantage was small, occurred in two of three folds, and was obtained after comparing many candidate feature sets. Although the search was chronological and bounded, selecting the best-performing specification can still adapt to the particular seasons used for development. The reversal in 2025/26 is consistent with a weak relationship that did not generalize, and illustrates why the final season had to remain outside feature and model selection.

The primary contrast also requires a qualification. The enhanced model added three player-performance variables and one expected-goals team-strength control. It therefore measured the combined contribution of that non-market bundle rather than player information in complete isolation. The development block checks provided additional evidence about the player variables, but the held-out primary comparison retained all four selected predictors. This does not weaken the finding that the complete enhanced model failed to outperform recalibration; it limits how specifically any hypothetical improvement could have been attributed.

8.2 Practical Meaning

The results show why closing probabilities should be treated as the principal benchmark when the claim concerns additional predictive information. The raw closing market outperformed the historical-frequency, Dixon–Coles, non-market logistic, and pooled LightGBM alternatives. Recalibration improved the raw market further, showing that the quoted probabilities were not beyond adjustment. Once that adjustment was made, however, the selected non-market features did not provide an additional held-out improvement.

For applied forecasting, the added features also carry a practical cost. They require collection and reconciliation of line-ups, player identities, minutes, positions, and event statistics, followed by chronological updating for every player. In the present setting, this extra data and modelling burden was not matched by a measurable improvement over the simpler recalibrated-market model. The result therefore provides no empirical reason to prefer the enhanced pooled model when reliable closing probabilities are already available.

Greater flexibility did not solve the problem. The LightGBM specifications could represent nonlinear relationships and interactions, but performed worse than the primary logistic models. Adding player features to Dixon–Coles also failed to improve its held-out result. Complexity can exploit additional patterns in development data, but it cannot create stable information that is absent from the predictors. In this application, the simpler market-based models generalized better.

These findings concern forecast quality rather than betting profitability. A model with lower log loss would not automatically identify a tradable strategy, because profitability also depends on the available price, bookmaker margin, limits, timing, and execution. Conversely, a model can have worse average probability scores yet identify a narrow set of mispriced fixtures. No staking rule or return-based strategy was evaluated here, so the results should not be described as evidence for or against betting profits.

The negative result remains useful. It rules out a plausible but costly feature design under a demanding evaluation and prevents a small development gain from being presented as a general forecasting improvement. It also leaves a reusable procedure for testing richer player representations against the same market-only baseline without changing the standard of evidence.

8.3 Differences Between Leagues

The country-level point estimates were heterogeneous. Portugal and the Netherlands favoured the enhanced pooled model, whereas Belgium, Scotland, and Turkey favoured recalibration alone. These directions could reflect differences in market efficiency, playing style, schedule structure, data quality, or the extent to which recent event statistics represent persistent ability. They could also arise from ordinary variation in samples of between 184 and 267 matches.

The uncertainty results support the cautious interpretation. Every country-specific interval crossed zero, including Portugal, which had the largest positive estimate, and Belgium, which had the largest negative estimate. The study therefore does not establish that the features are useful in one country and harmful in another. Such a claim would require additional seasons evaluated under a specification fixed before those observations were seen.

The league-specific models offer a related but incomplete indication of heterogeneity. Their enhanced version achieved the lowest descriptive log loss in the final model table, improving slightly on league-specific recalibration. This may suggest that the relationship between market probabilities and the selected features differs across competitions and is weakened when a single set of feature coefficients is shared. However, five independently fitted models have greater flexibility, the result was secondary rather than the predeclared primary test, and its advantage was not consistent across all scoring measures. It should consequently be treated as a hypothesis for future evaluation rather than a reason to revise the pooled conclusion after observing the holdout.

8.4 Relationship to Previous Research

The strength of the closing benchmark is consistent with earlier evidence that bookmaker odds are difficult for statistical football models to outperform. Forrest et al. (2005) found that a detailed statistical model did not improve on odds-setters' forecasts in English football, while Štrumbelj and Robnik-Šikonja (2010) documented the forecasting quality of online bookmaker odds across several European leagues. The present study extends this line of evidence to five mid-sized European competitions and to a direct test of announced-line-up performance information.

The distinction between raw and recalibrated probabilities is also important. Štrumbelj (2014) showed that the method used to convert odds into probabilities affects forecast assessment, and Angelini and De Angelis (2019) reported that football betting-market efficiency can vary across settings. In the current results, recalibration improved the normalized closing probabilities. A comparison against the raw market alone would therefore have mixed the contribution of player information with the contribution of correcting market probability biases. Using the recalibrated market made the incremental test more demanding and more directly connected to the research question.

The findings do not contradict studies showing that player ratings improve team-based models. Arntzen and Hvattum (2021), for example, found value in combining team and player ratings. The small development improvement from adding players to Dixon–Coles is compatible with the idea that player information can help a model based on team history. The present paper applies a different and stronger standard: whether player information improves predictions after conditioning on closing market probabilities. Success against a team model does not imply success against an information-aggregating market.

The weak contribution of recent event summaries is closer to the findings of Gelade and Hvattum (2020), who reported only marginal predictive gains from event-level performance statistics when combined with player ratings. It also clarifies the difference from Holmes and McHale (2024). Their approach modelled changing underlying player ability, whereas the present study aggregated recently observed output. Latent ability and short-term recorded form are related but distinct quantities: the former attempts to estimate persistent quality, while the latter is directly observable but more exposed to temporary match context.

Overall, the study reinforces the importance of matching the benchmark to the claim. The player variables contained some development-sample signal and performed better than shuffled values, but that was not sufficient to produce a reliable improvement over recalibrated closing probabilities in the final pooled evaluation. Closing markets appear to absorb most of the useful information represented by these features, while the remaining contribution is small relative to season-to-season variation. Richer player representations may still prove useful, but they should be tested using the same chronological, market-adjusted, and independently held-out design.

9 Limitations

The study tested a deliberately specific form of player information. Recent key passes, shots, and defensive actions described observed output over each starter’s previous five appearances, but did not represent every aspect of the player’s contribution. The features did not directly measure underlying ability, injuries, physical condition, tactical instructions, expected minutes, or the quality of substitutes. They also reduced a starting eleven to aggregate totals and therefore captured partnerships, positional balance, and interactions between players only indirectly. The findings apply to these recorded performance summaries rather than to player information in general.

The primary held-out comparison was not purely player-only. The enhanced model added three player-performance variables and one expected-goals team-strength control to the recalibrated market. Its result therefore measured the combined contribution of that four-feature

bundle. The development robustness analysis removed feature blocks and provided additional evidence about the player variables, but it was conducted before the final evaluation and on an earlier five-feature candidate. Because a player-only version was not designated as a separate primary model for 2025/26, the final result cannot establish how the three player variables would have performed without the team-strength control. It establishes that the complete predeclared enhanced specification did not improve the pooled market forecast.

The five-appearance window involved a trade-off between recency and stability. It could respond quickly to a change in recorded performance, but it also contained relatively few observations. Event rates depend on opponent quality, position, tactics, score state, and playing time. Per-90 conversion and the candidate opponent-adjusted features addressed some of this context, but could not turn short-term event counts into a direct measure of persistent player quality. A longer history or a dynamic player-strength model may provide a more stable signal, although it would examine a related rather than identical question.

The geographical and temporal scope also limits generalization. The final evidence came from one season in five national competitions, with between 184 and 267 eligible matches per country. These samples were sufficient for the combined analysis but produced wide country-level intervals. Greece was excluded because its first two seasons did not meet the common starting-line-up coverage requirement. The results should therefore not be assumed to apply to Greece, other mid-sized European leagues, the Big Five, lower divisions, or international football. Further seasons evaluated without revising the specification would be needed to determine whether the observed pattern remains stable over time.

The strict complete-case rules improved comparability but reduced representativeness. A fixture was retained only when the scores agreed across sources, valid closing odds were available, both starting elevens contained exactly eleven identified players, and the required player records met the coverage rules. Matches with incomplete line-ups or weaker data coverage were excluded. If those fixtures differed systematically from the retained sample, the reported performance may not describe every match played in the five leagues. The coverage and exclusion tables make this selection visible but cannot remove the resulting limitation.

Player histories were also restricted to appearances in the player’s current league. This prevented performances from competitions with different playing and measurement environments from being treated as directly equivalent. It also discarded potentially useful earlier information for transferred players, leaving them with limited local history after a move. A future extension could carry information between countries after estimating an explicit adjustment for league strength and data comparability.

Closing odds are an unusually information-rich benchmark. They may reflect line-ups, injuries, transfers, recent performances, private information, and judgements not present in the research dataset. The study could determine whether the engineered variables improved predictions conditional on those prices, but it could not identify exactly which pieces of player information the market had already incorporated. Failure to add predictive value does not mean that the underlying information was irrelevant to the formation of the market probabilities.

The model suite was broad enough to compare linear, tree-based, and score-distribution approaches, but it was not exhaustive. Detailed dynamic player-ability models, tactical representations, explicit absence models, and networks of player interactions were outside the

declared scope. Their omission does not alter the findings for the evaluated specifications, but it prevents a claim that no possible model could extract value from player data. Furthermore, although the untouched season protected the main result from continued feature selection, one final season cannot represent every future market or competitive regime.

Finally, the study assessed probability forecasts rather than betting returns. It did not model the price available after generating a prediction, bookmaker limits, transaction constraints, market movement, or a staking rule. Better log loss would not by itself establish a profitable strategy, and the absence of an average forecasting improvement does not rule out isolated pricing errors. The results should therefore be interpreted as evidence about probabilistic prediction, not as a direct evaluation of betting profitability or a causal effect of player performance on match outcomes.

These limitations narrow rather than remove the contribution of the study. The results address whether a specific and reproducible set of recent starter-performance summaries improved forecasts beyond closing odds under a chronological multi-league design. They do not establish that all forms of player information are uninformative or that such information could never support a useful application.

10 Conclusion

This study examined whether recent recorded performances from announced starting players improved three-way football match forecasts beyond recalibrated closing bookmaker probabilities. The analysis covered 7,801 eligible matches from Belgium, the Netherlands, Portugal, Scotland, and Turkey. Feature and model decisions were made through chronological development experiments covering 2020/21–2024/25, after which the complete specification was evaluated once on 1,163 held-out matches from 2025/26.

Development results suggested a small but inconsistent advantage from the selected enhanced specification. That advantage did not persist in the final pooled evaluation. The enhanced model recorded an equal-league log loss of 0.9580, compared with 0.9567 for the recalibrated market, representing a deterioration of 0.143%. Brier score and RPS showed the same direction. Country-level estimates were positive in Portugal and the Netherlands and negative in Belgium, Scotland, and Turkey, but every uncertainty interval included zero.

These results provide no reliable evidence that the selected combination of recent starter-performance summaries and team-strength context improved on recalibrated closing probabilities. They also show why a player-enhanced model should be compared with a trained market-only baseline rather than only with historical frequencies, team-strength models, or raw odds. Recalibration itself can produce an apparent improvement that might otherwise be attributed incorrectly to the added information.

The findings do not imply that every form of player information lacks predictive value. More stable player-strength estimates, richer representations of tactical roles and player interactions, or adjusted histories for players moving between leagues may capture information that the tested rolling summaries missed. Such extensions should retain chronological feature construction, identical-match comparisons, multi-league reporting, and an untouched final evaluation. Within the present design, however, the added data and model complexity were not matched by a dependable improvement over the closing market.

AI Usage Disclosure

During the preparation of this work, the author used ChatGPT within the Codex environment to assist with selected coding tasks, code documentation, and streamlining the development workflow. Grammarly was subsequently used to review the manuscript for grammar, spelling, and writing style. All outputs generated with the assistance of these tools were critically reviewed, verified, and edited by the author. The author takes full responsibility for the accuracy, integrity, and final content of the manuscript and the accompanying code.

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A Data Quality and Sample Coverage

The common eligibility procedure was applied independently inside each league before the country tables were combined. A match entered the modelling sample only when it was a completed top-division fixture, appeared in both match sources with an agreeing score, had valid closing prices for all three outcomes, contained player records, and supplied exactly eleven identified starters for each team with valid minutes and team assignments. The same requirements were used in development and in the held-out season.

Greece was not included because its starting-line-up coverage in the first two seasons failed the declared coverage requirement. The exclusion was made before the five-league model comparisons and was not determined by forecasting performance. Portugal’s reduced 2023/24 sample was retained under the common rules rather than replaced by a country-specific exception.

Table 8: Eligible observations used for final training and evaluation.

League	2020/21–2024/25	2025/26	Total
Belgium	1,494	267	1,761
Netherlands	1,459	255	1,714
Portugal	1,000	211	1,211
Scotland	1,121	184	1,305
Turkey	1,564	246	1,810
Total	6,638	1,163	7,801

B Full Model Specifications

Table 9: Information and estimation scope of the evaluated models.

Model	Training scope	Information used
Historical frequency	Shared and separate	Earlier league-specific H/D/A frequencies
Raw closing market	None	Margin-removed closing home, draw, and away prices
Recalibrated market	Shared and separate	Closing-market log ratios and league indicators
Non-market logistic	Shared and separate	Selected chronological player and team variables, without odds
Enhanced market logistic	Shared and separate	Recalibrated-market inputs plus four selected non-market variables
Player-feature LightGBM	Shared and separate	Selected home-minus-away player and team variables
Expanded LightGBM	Shared and separate	Selected home and away variables retained separately
Dixon–Coles	Separate by league	Earlier scores, time-weighted team attack and defence, and home advantage
Dixon–Coles with players	Separate by league	Dixon–Coles structure plus selected announced-line-up variables

Table 10: Fixed estimation and evaluation settings.

Component	Fixed setting
Logistic regression	Standardized inputs; L_2 regularization with $C = 1$; limited-memory BFGS; maximum 2,000 iterations
LightGBM	100 trees; learning rate 0.03; maximum depth 3; 7 leaves; minimum 30 observations per leaf; L_2 penalty 1.0
Shared training	Each league assigned equal total training weight
Dixon–Coles	Separate model per league; older matches exponentially downweighted with a 365-day half-life
Model seed	42
Primary score	Equal-league log loss
Secondary scores	Brier score and ranked probability score
Uncertainty	10,000 paired match-week bootstrap repetitions; seed 42
Post-hoc calibration	None in the final evaluation

C Selected Features

The entries below are the exact names loaded by the final evaluators. They were selected entirely from the chronological development folds. The shared and separate versions of a model used the same feature list so that training scope, rather than input availability, was the relevant difference.

Table 11: Final selected features by model family.

Model family	Selected feature
Non-market logistic	elo_rating opponent_elo_rating elo_difference defence_goal_rate_ewm expected_goals_strength key_passes_per90_sum_5 adjusted_defensive_actions_lineup_mean_5 adjusted_key_passes_lineup_std_across_starters_5
Enhanced market logistic	key_passes_per90_sum_5 shots_per90_sum_5 defensive_actions_per90_sum_5 expected_goals_strength
Player-feature LightGBM	key_passes_per90_sum_5 shots_per90_sum_5 defensive_actions_per90_sum_5 recent_minutes_sum_5 rating_mean_5 starters_without_history starters_without_full_window elo_rating opponent_elo_rating elo_difference elo_expected_result prior_team_matches attack_goal_rate_ewm defence_goal_rate_ewm expected_goals_strength key_passes_per90_lineup_mean_ewm adjusted_defensive_actions_lineup_mean_1 rating_lineup_mean_3 adjusted_rating_lineup_std_across_starters_5
Expanded LightGBM	elo_rating opponent_elo_rating elo_difference elo_expected_result prior_team_matches

Model family	Selected feature
	attack_goal_rate_ewm key_passes_per90_sum_5 adjusted_key_passes_lineup_mean_1 gk_saves_per90_5 adjusted_key_passes_lineup_mean_ewm
Dixon–Coles with players	key_passes_per90_sum_5 recent_minutes_sum_5 rating_mean_5 starters_without_history mid_rating_mean_5 fwd_adjusted_shots_5 gk_saves_per90_5

D Exhaustive Feature-Group Searches

All nonempty combinations of the candidate feature groups were considered for each applicable model family, producing 63 candidates when all six groups were available. Black marks in the left panels indicate included groups. The right panels order the combinations by mean development log loss and show their distance from the best candidate in the same family. The orange band marks the predeclared 0.005 log-loss tolerance within which the smaller candidate was preferred. “Recent player output” denotes the core five-appearance summaries of observed player actions; it replaces the older internal label used during early code development.

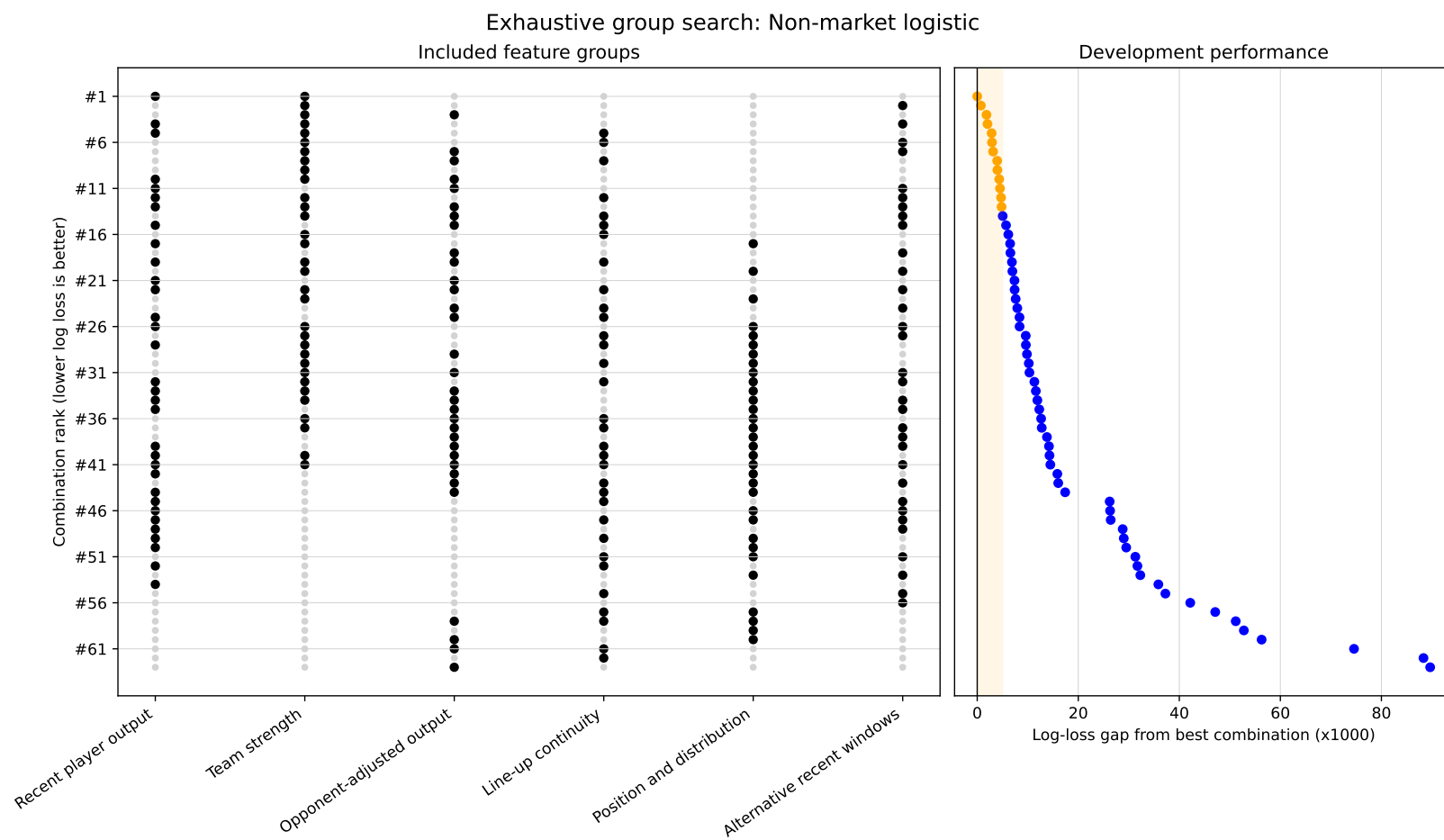


Figure 12: Exhaustive group search for the non-market logistic model.

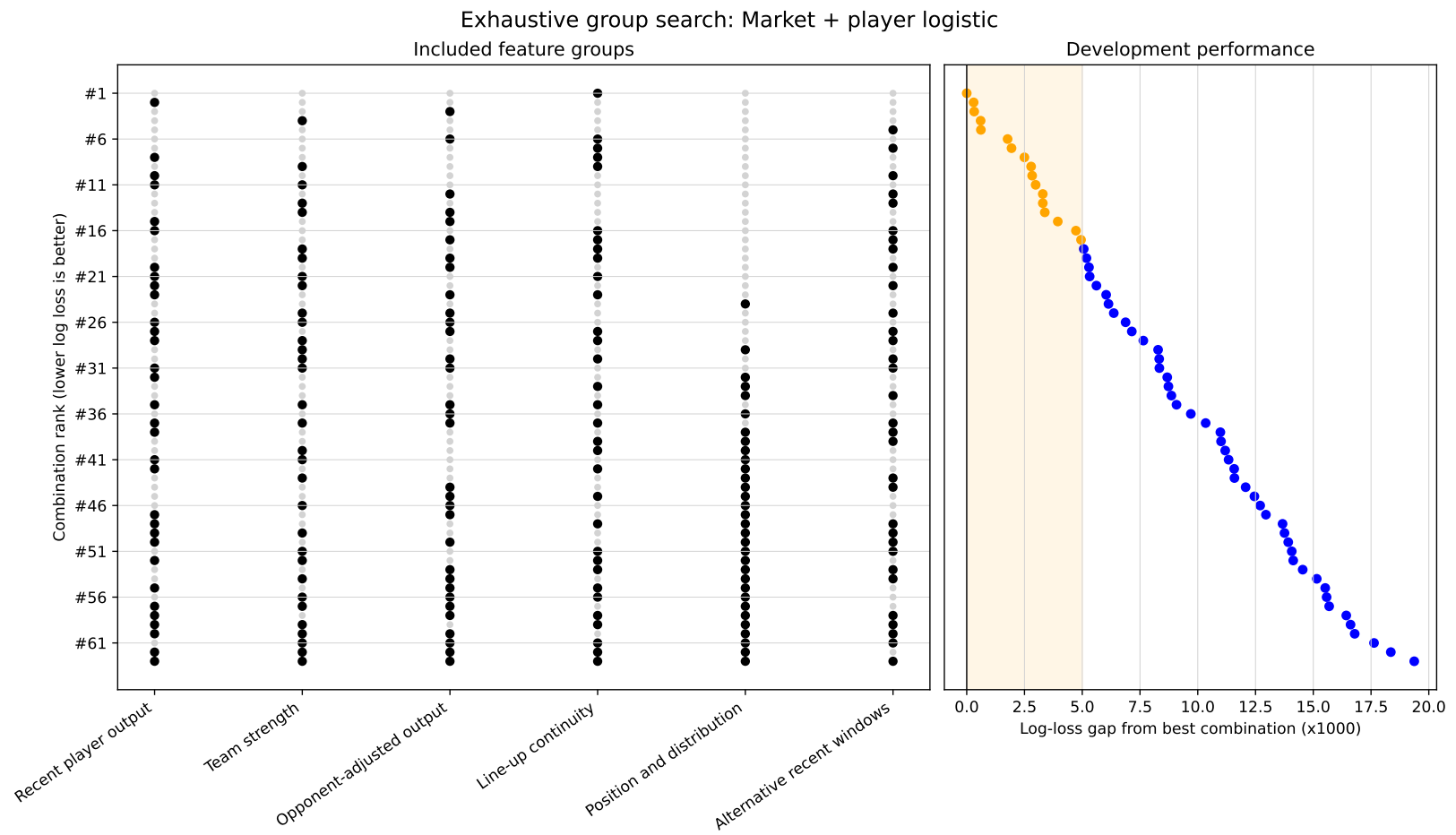


Figure 13: Exhaustive group search for the market-plus-player logistic model.

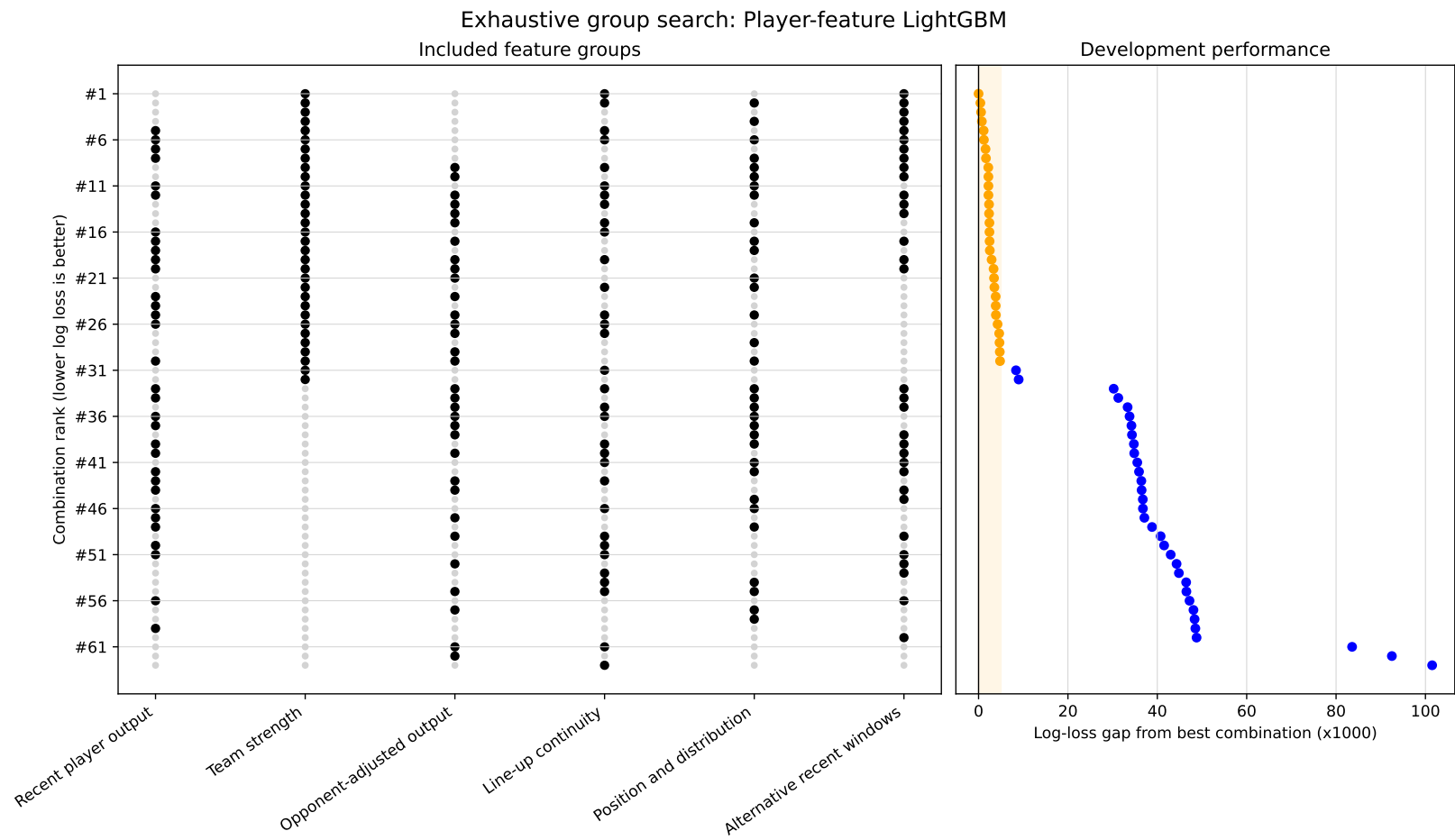


Figure 14: Exhaustive group search for the player-feature LightGBM model.

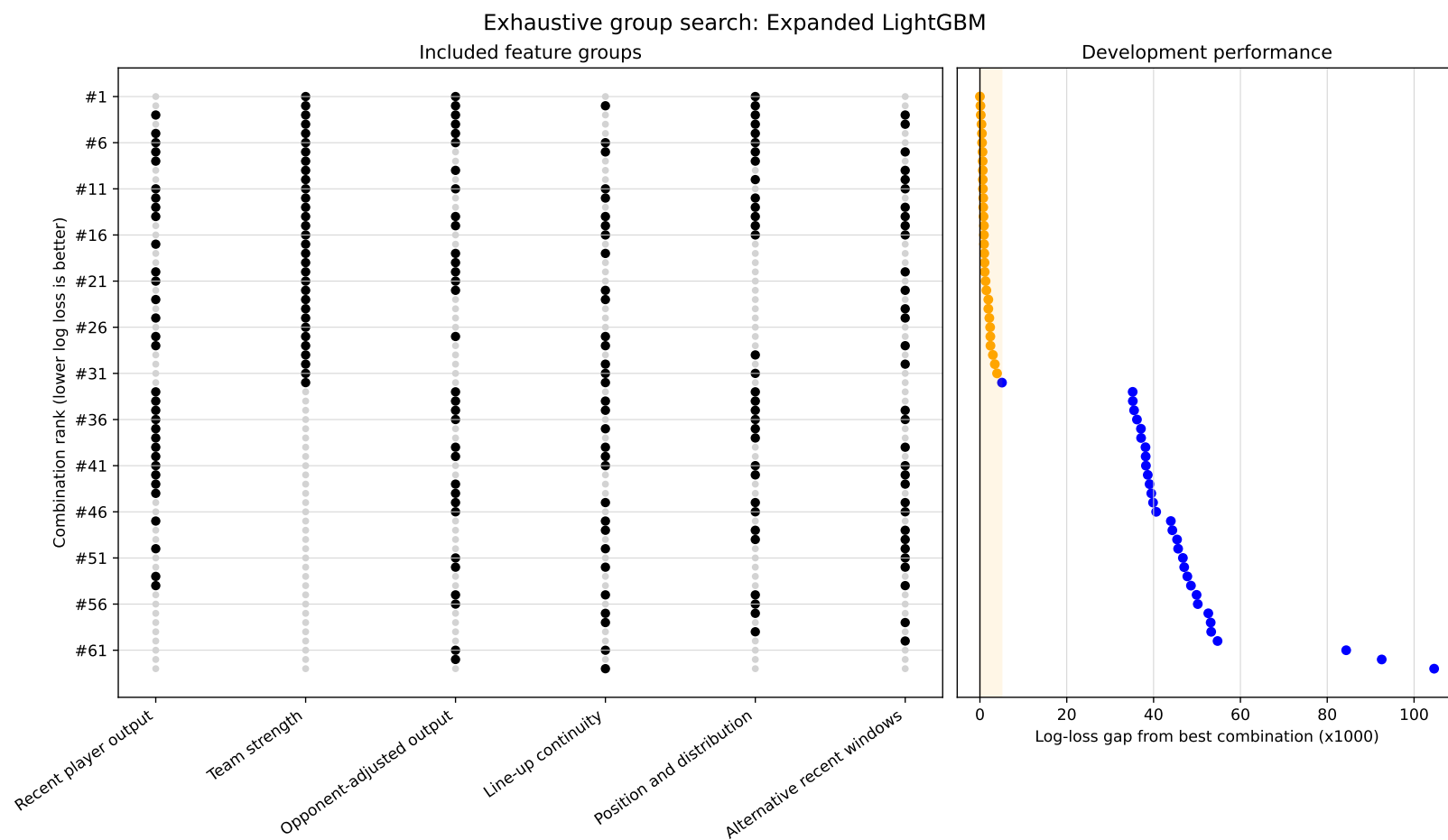


Figure 15: Exhaustive group search for the expanded LightGBM model.

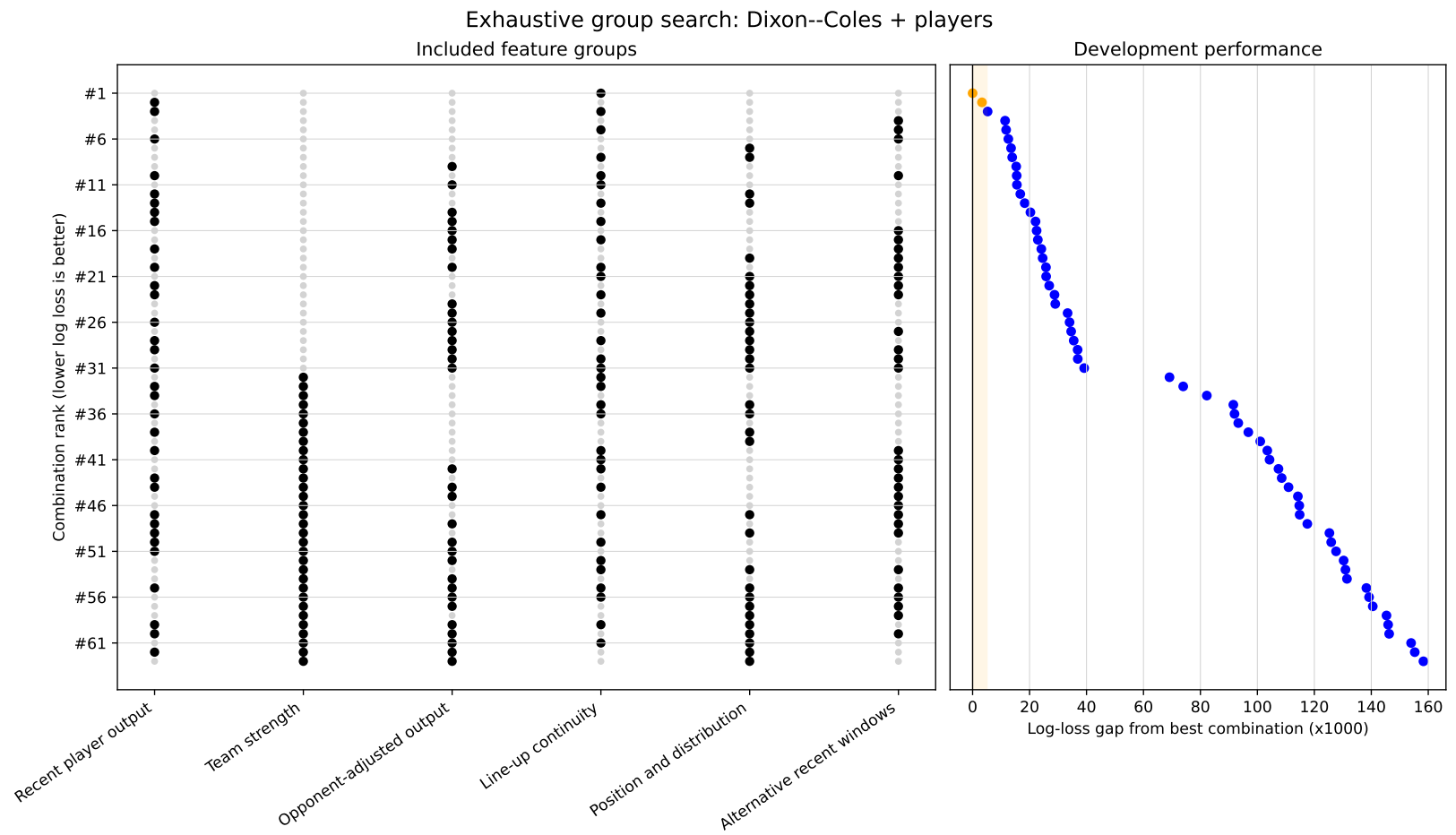


Figure 16: Exhaustive group search for Dixon–Coles with player information.

E Additional Development and Diagnostic Results

Figure 17 reports the focused development checks relative to the five-feature precursor to the primary model. Negative values mean that the alternative reduced log loss. Removing the mean player rating was the only tested removal that improved the precursor and therefore produced the four-feature final specification. The remaining feature-block removals, the market-only comparison, and the player shuffle all increased loss, although the changes were small.

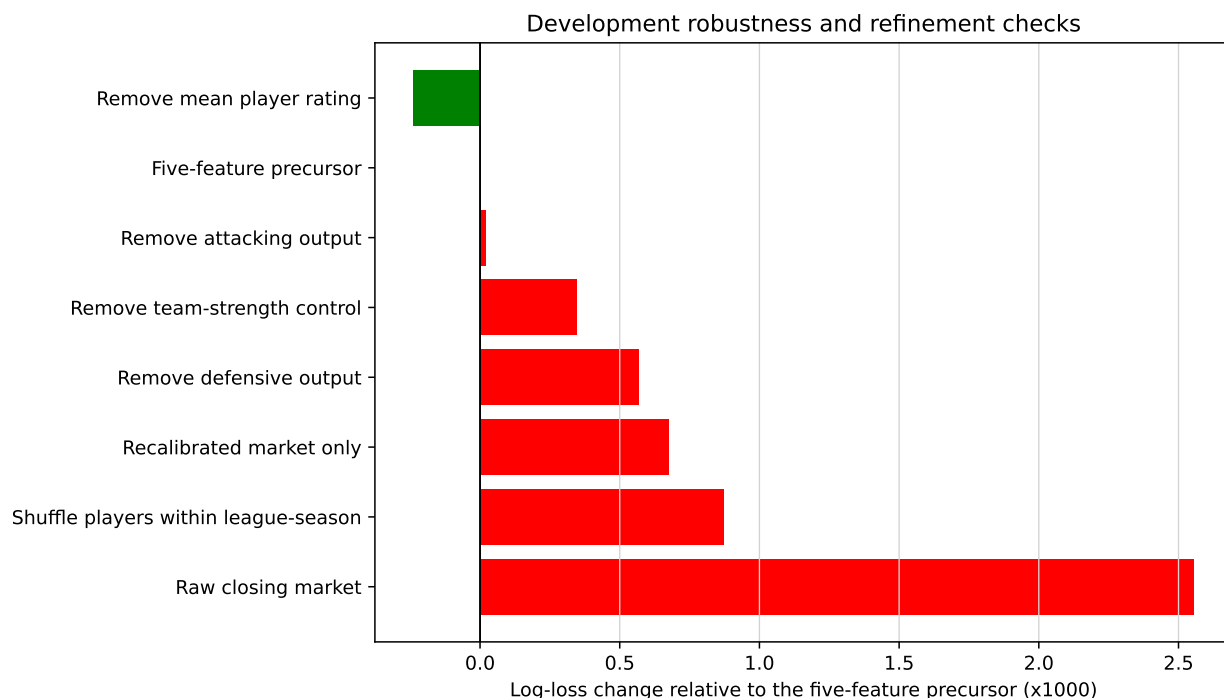


Figure 17: Development robustness checks relative to the five-feature precursor. Values are log-loss changes multiplied by 1,000.

Figure 18 compares development-period probability reliability for the raw closing market and the shared and separately trained enhanced models. It is diagnostic rather than the primary evaluation: the final research claim rests on proper scoring rules and the paired comparison against the recalibrated market.

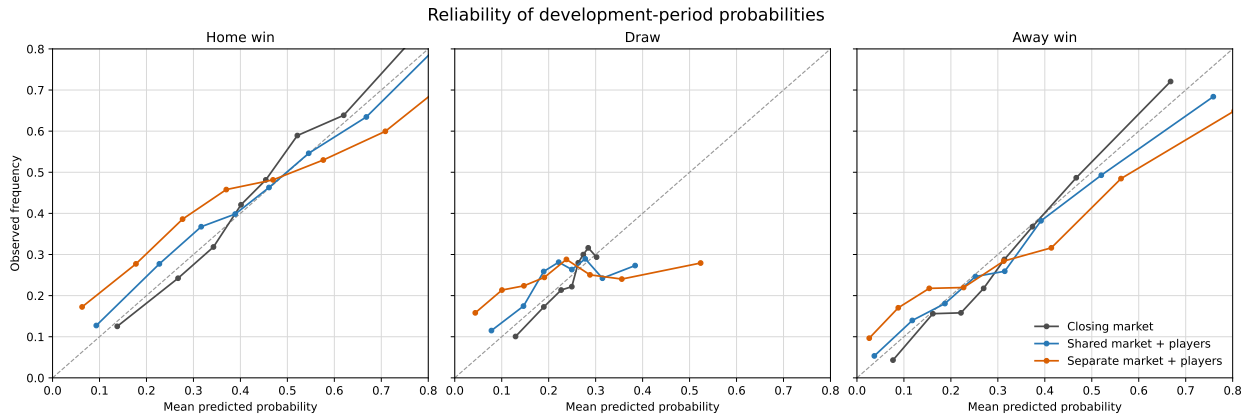


Figure 18: Development-period reliability by outcome. Points compare mean predicted probabilities with observed frequencies in probability bins.

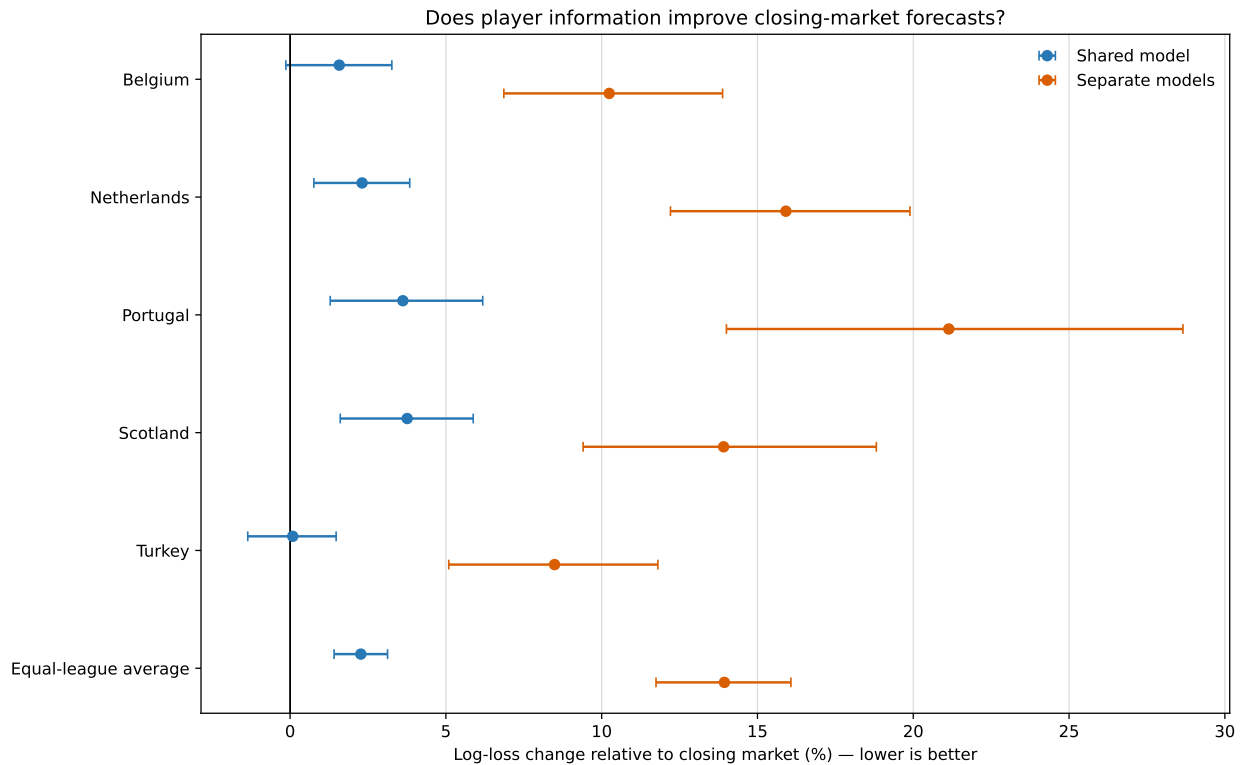


Figure 19: Development-period changes in log loss relative to the raw closing market. The figure is contextual and does not replace the recalibrated-market primary comparison.

F Reproducibility Details

The final-evaluation settings were stored separately from the development evaluator. They fixed the five training seasons, the 2025/26 test season, the five included leagues, the complete model list, the absence of post-hoc calibration, the scoring and aggregation rules, and the uncertainty procedure. The selected-feature file was checked against SHA-256 value `2f3f3a757a26bcd3df93eb798ae8dd8135d82bb40ab4c695a39f0846c4655df4` before final fitting.

The final evaluator saved match-level probabilities for every comparator before any summary measure was calculated. Equal-league, match-weighted, and country-level scores were all derived from these common predictions. The paired bootstrap resampled match weeks within league and season so that every model was evaluated on the same resampled observations.

The publication-specific figures in this paper can be rebuilt without fitting any model by running:

```
python multi_league_research/visuals/scripts/paper_figures.py
```

That command reads the saved final-report, feature-selection, and robustness tables and writes both PDF and PNG versions to `artifacts/paper/figures`. The manuscript itself is contained in `artifacts/paper/main.tex`; the final settings and selected inputs are in `multi_league_research/config/final_evaluation.json` and `multi_league_research/config/selected_features.csv`, respectively.